Nadel · payment accounting for garden cooperatives

Board guide

How the chairman, the treasurer and the electrician work in the system, and the rules it calculates by: tariffs, readings, charges, losses, water — with worked examples.

Reader	Chairman, treasurer, electrician, other board members
System state	Phases 1–2: part I — working in the system, part II — how the system calculates
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1 The system, roles and signing in

Nadel keeps the cooperative's records: who the members are, which plots they own, what is charged and at which tariffs, how much electricity and water was used. Charges, payments, invoices and the map come next.

Who sees what

Everyone signs in with their own account. What a person sees and may do depends on their **roles**, and roles are made of **permissions** (chapter 3). Every cooperative has three base roles:

Role	May do
Chairman	Everything: members, roles, plots, owners, tariffs, meters, readings, import, settings, audit log. The chairman's rights cannot be reduced, and a cooperative always keeps at least one chairman.
Treasurer	Views members, plots, tariffs, meters and the audit log. Changes nothing until the chairman adds permissions.
Member	Own data only: own plots and their tariffs, own meters, submitting readings.

The chairman can create custom roles. For example the role "Electrician" has only two permissions, "View meters" and "Enter readings"; a person with it sees only "Readings" and "Meters" in the menu.

Signing in

1. Open the system's address and press "Sign in".
2. Enter your email and password.
3. The system finds you among the members by email. If your member card has no email or a different one, you see "Account not linked": ask the chairman to correct the email on your card.

> NOTE
>
> For now the sign-in account (email and password) is created by the platform administrator. Self sign-up and email invitations come later.

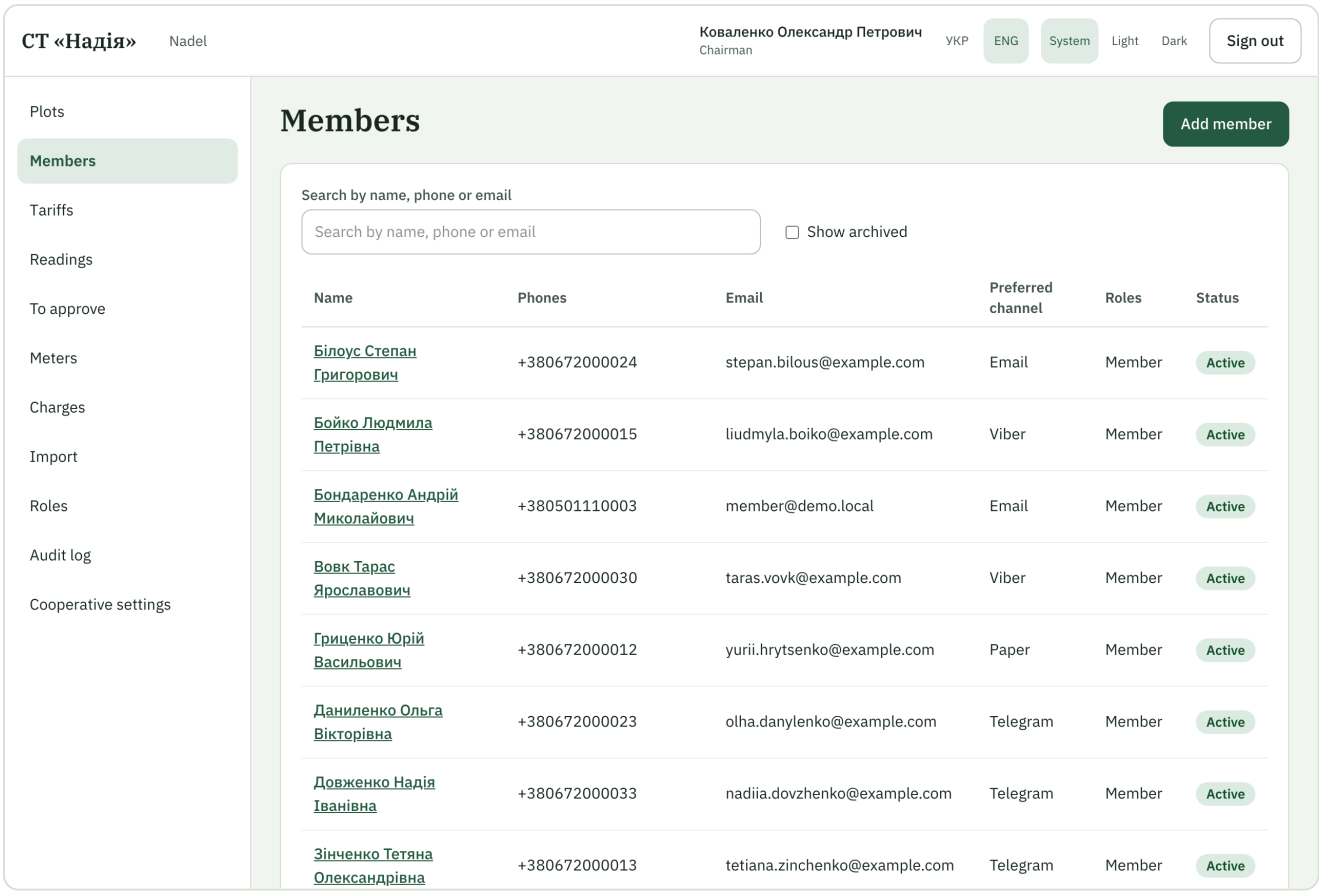
Language and theme

At the top right: the language switch **УКР / ENG** and the theme **System / Light / Dark**. The choice is remembered in this browser. The interface always shows exactly one language: with English selected, every name, hint and message is English.

At the top left is your cooperative's name, below it the menu. On a phone the menu hides behind the "Menu" button.

2 Members

"Members" lists every member, with search by last name, phone or email.



The members list

Adding a member

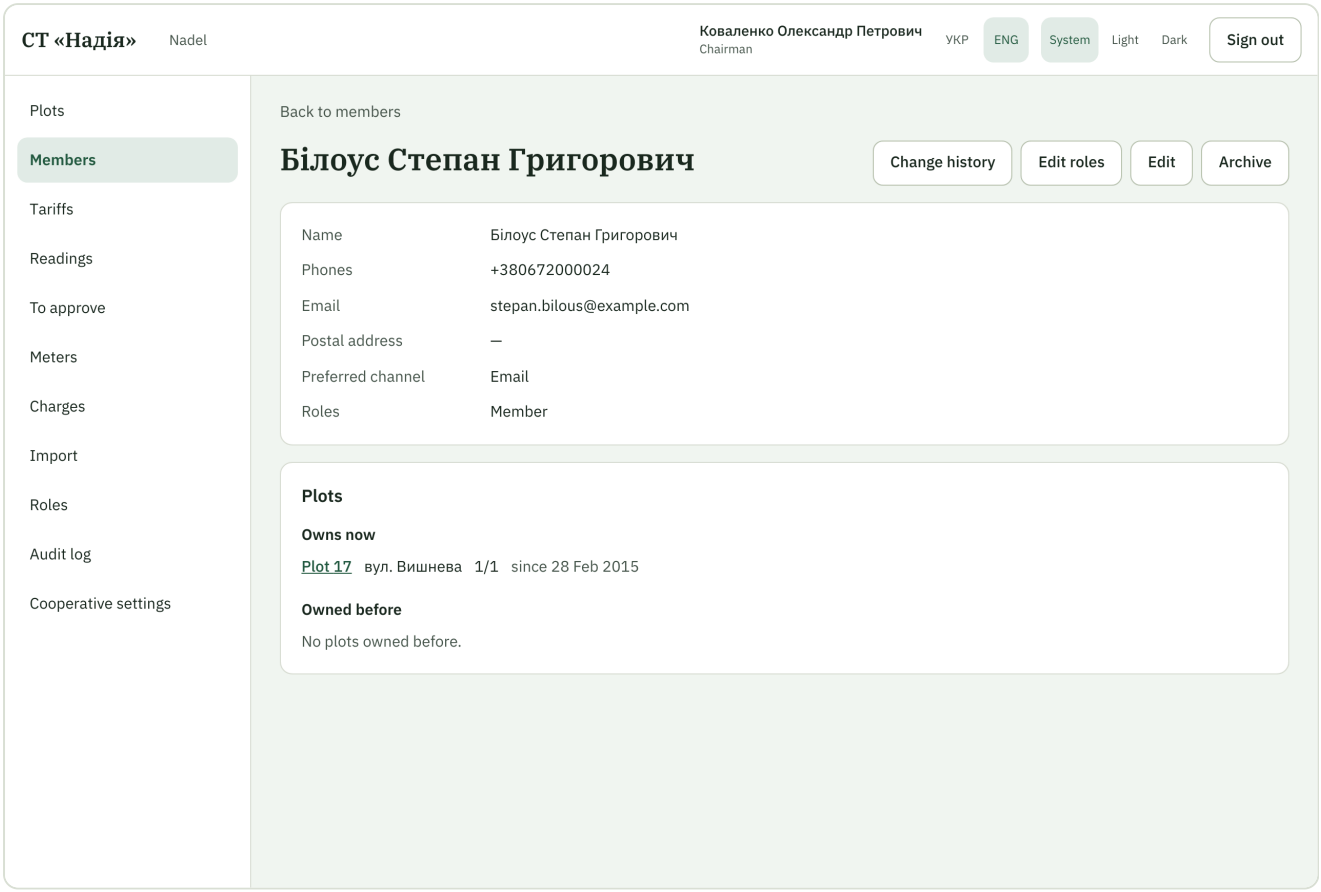
1. Press "Add member".
2. Fill in the last and first name (required), patronymic, phones, email, postal address.
3. Choose how invoices are delivered: email, Telegram, Viber or paper. The matching contact is required.
4. Press "Save". A phone typed as `0XXXXXXXXX` is stored as `+380XXXXXXXXX`.

NOTES AND PERSONAL DATA

Notes are for working information about a member: how best to reach them, arrangements, relatives' contacts. Only people with "View members" see them. The system has no fields for tax numbers (РНОКПП), passport data or dates of birth — the cooperative does not need them for its records.

The member card

Contacts, roles, owned plots and notes. Notes are visible only to people with "View members".



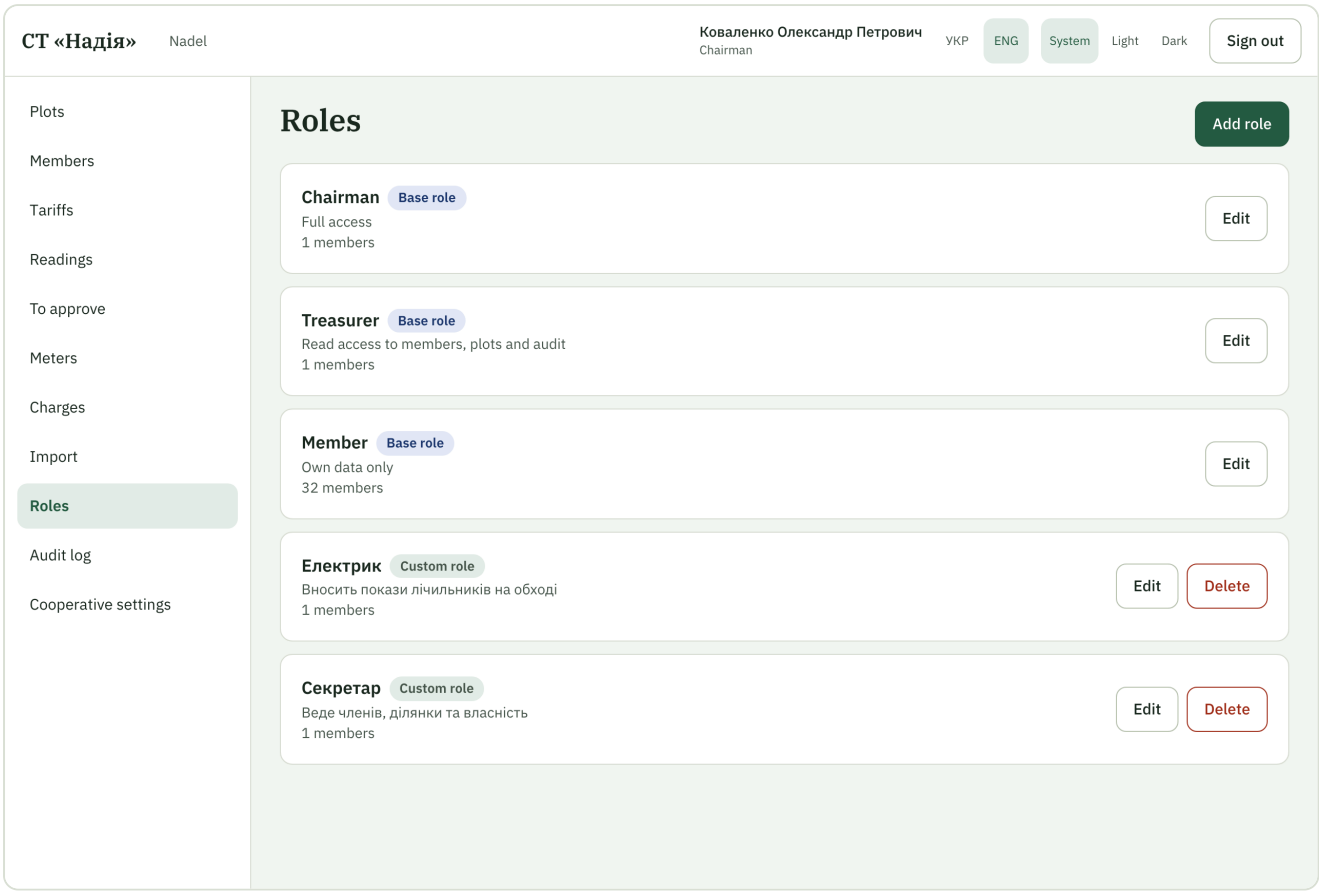
A member card

Archiving

A member is archived, not deleted: hidden from the list (visible with "Show archived"), unable to sign in, history kept. "Restore" brings them back. A current plot owner cannot be archived until the plot changes hands, nor the last chairman until another is appointed.

3 Roles and permissions

In "Roles" the chairman (or anyone with "Manage roles") decides who may do what.



Base and custom roles

- **Base roles** (Chairman, Treasurer, Member) cannot be deleted but can be renamed; the new name and description then show in every language. "Restore standard name" undoes it.
- **Custom roles**: "Add role", a name, a description and any set of permissions. A person may hold several roles; permissions add up.
- A role still assigned to someone cannot be deleted.

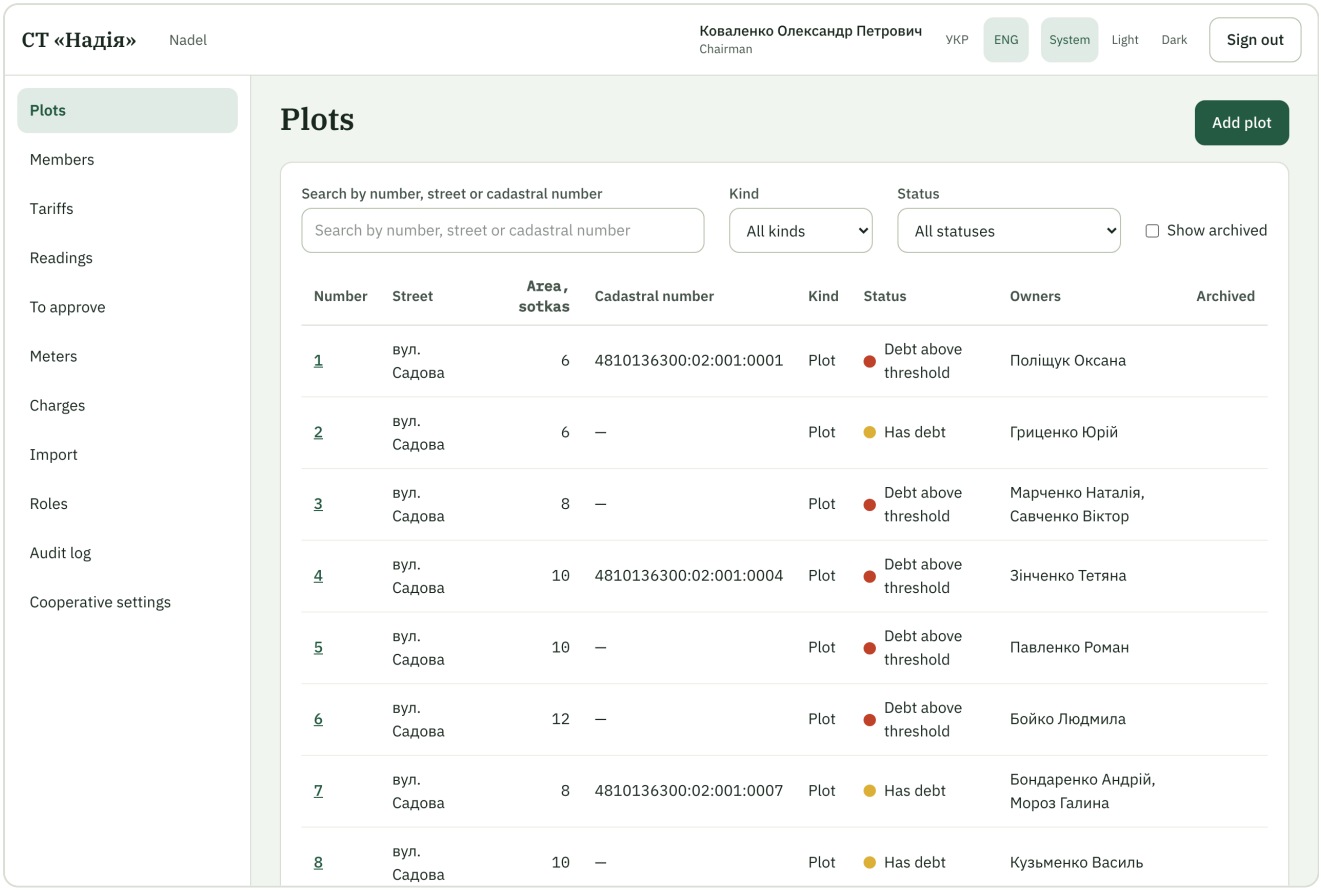
Permissions

Permission	Allows
View / edit members	See all members and contacts; add, change, archive
Manage roles	Create roles, change permissions, assign roles
View / edit plots	See every plot in detail; add, change, archive
Edit ownership	Add, change and correct owners

Permission	Allows
Run import	Upload data from Excel
Manage settings	Change the cooperative settings
View / manage tariffs	See tariffs; create charge types, tariffs and versions
View / manage meters	See the register and the tree; register, replace, remove meters
Enter readings	Enter, correct and delete readings (the electrician's round)
Approve readings	Approve or reject readings submitted by members
View audit log	Read the audit log

4 Plots

"Plots" is the register of every plot, with search by number, street or cadastral number and filters by kind and status.



The plot register

What is kept about a plot

Number — Required, unique in the cooperative: `12`, `12a`, `R-1`. Natural order: 2 before 10, 12a after 12.

Kind — **Plot** or **Common land** (roads, shared areas). Common land has no owners, tariffs or plot meters.

Area — In sotkas or m², entered in either unit (1 sotka = 100 m²).

Cadastral number — Optional, format `NNNNNNNNNN:NN:NNN:NNNN`.

Building — Whether a building stands on the plot. Reference only; it never affects charges.

Notes — Visible only with "View plots".

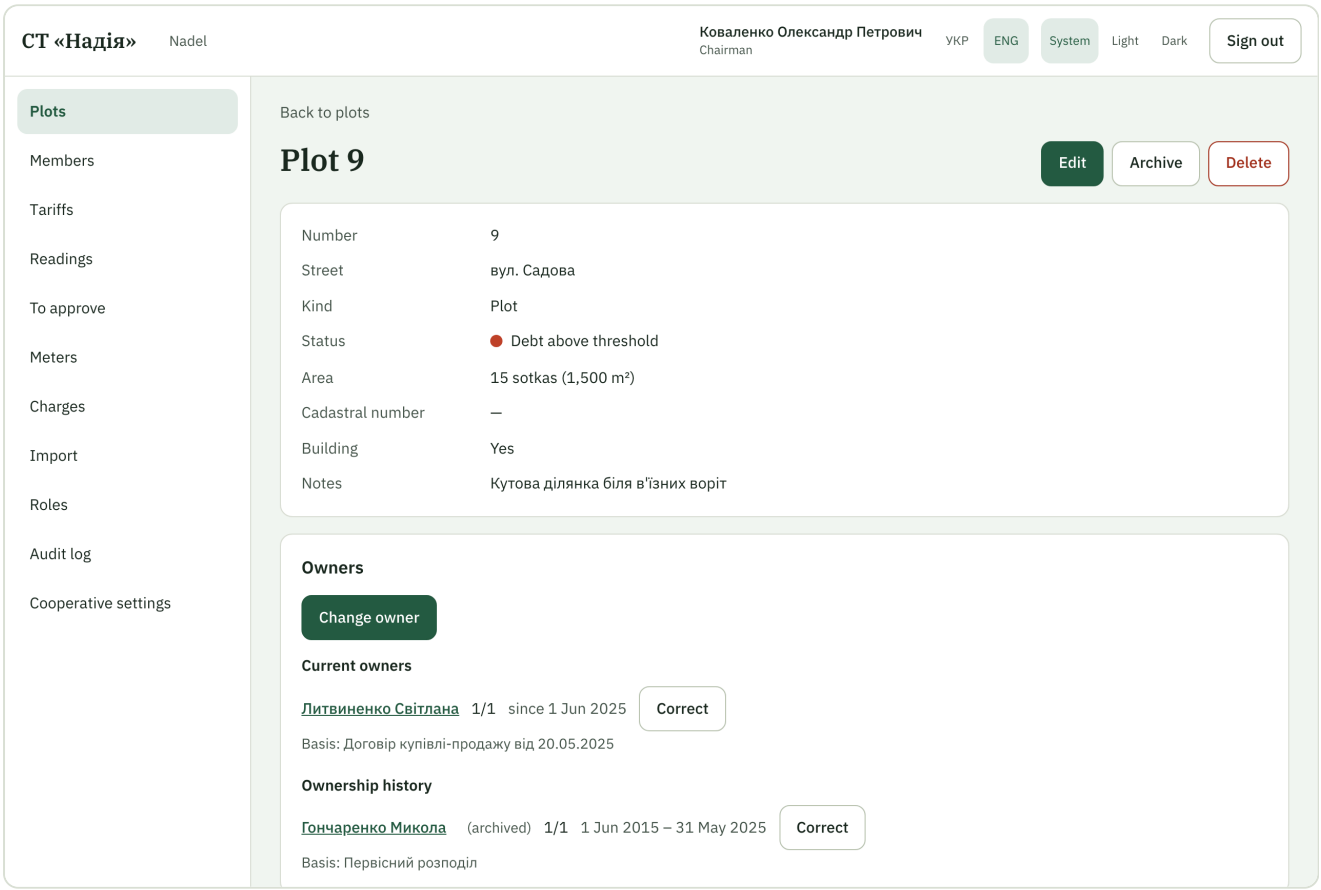
Plot status (colour)

Colour	Meaning
Green	No debt
Yellow	Has a debt
Red	Debt over the threshold (amount or age set by the chairman)
Grey	A plot without an owner, or common land

Until charges exist, a plot with an owner is green. Colours start to show debts with the charges (chapter 17).

The plot card

All the plot's data, owners with history, tariffs for a chosen month, meters.



Plot 9: owners, tariffs, meters

"Archive" hides a plot from the list and keeps its history. Only a plot created by mistake that never had an owner can be deleted.

5 Plot owners

Owners are kept on the plot card. A plot can have several co-owners with shares (`1/2`, `1/3`); the shares always add up to one. One member may own several plots, each with its own account.

Change of owner (sale, inheritance)

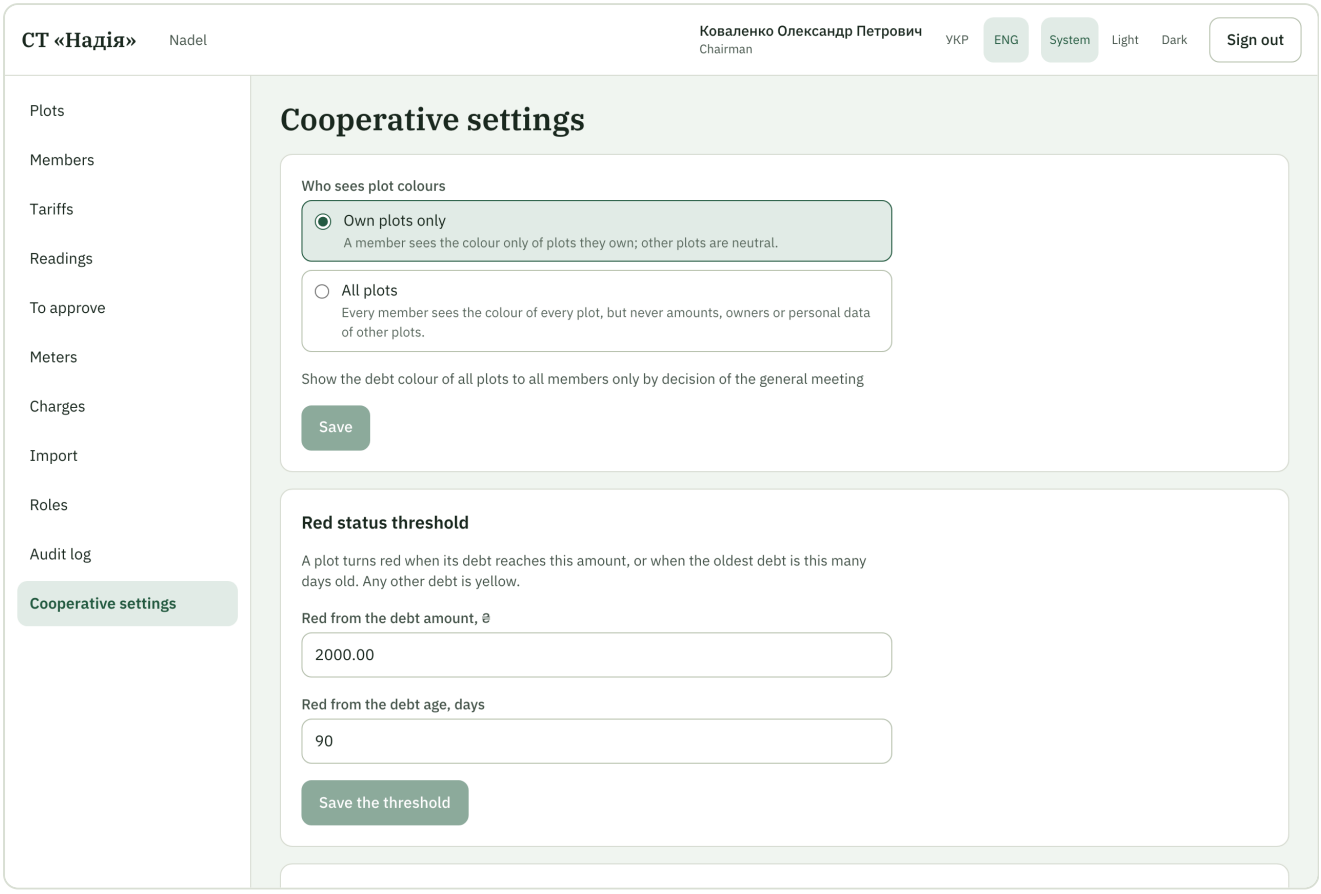
1. On the plot card press "Change owner".
2. Enter the date and the new owners with shares. If a new owner is not a member yet, use "Add member" right in the form.
3. Enter the basis (purchase contract, inheritance, court decision) and save.

The previous owners stay in the history up to the day before. **A debt belongs to the plot, not the person:** after a sale it stays on the plot.

Correcting a mistake

A wrong share, date or person in a record is fixed with "Correct" next to the record. The old values stay in the audit log. Ownership records are never deleted.

6 Cooperative settings



Cooperative settings

Who sees plot colours

- **Own plots only** (default): a member sees the colour of their own plots only; other plots are neutral.
- **All plots**: everyone sees every plot's colour, but never the amounts, owners or personal data of other plots.

> **WHY "OWN PLOTS ONLY" IS THE DEFAULT**
>
> Neighbours know whose plot is whose, so the colour effectively names the debtor. Switch to "All plots" only after a general meeting decision. The change is written to the audit log.

Meter tree

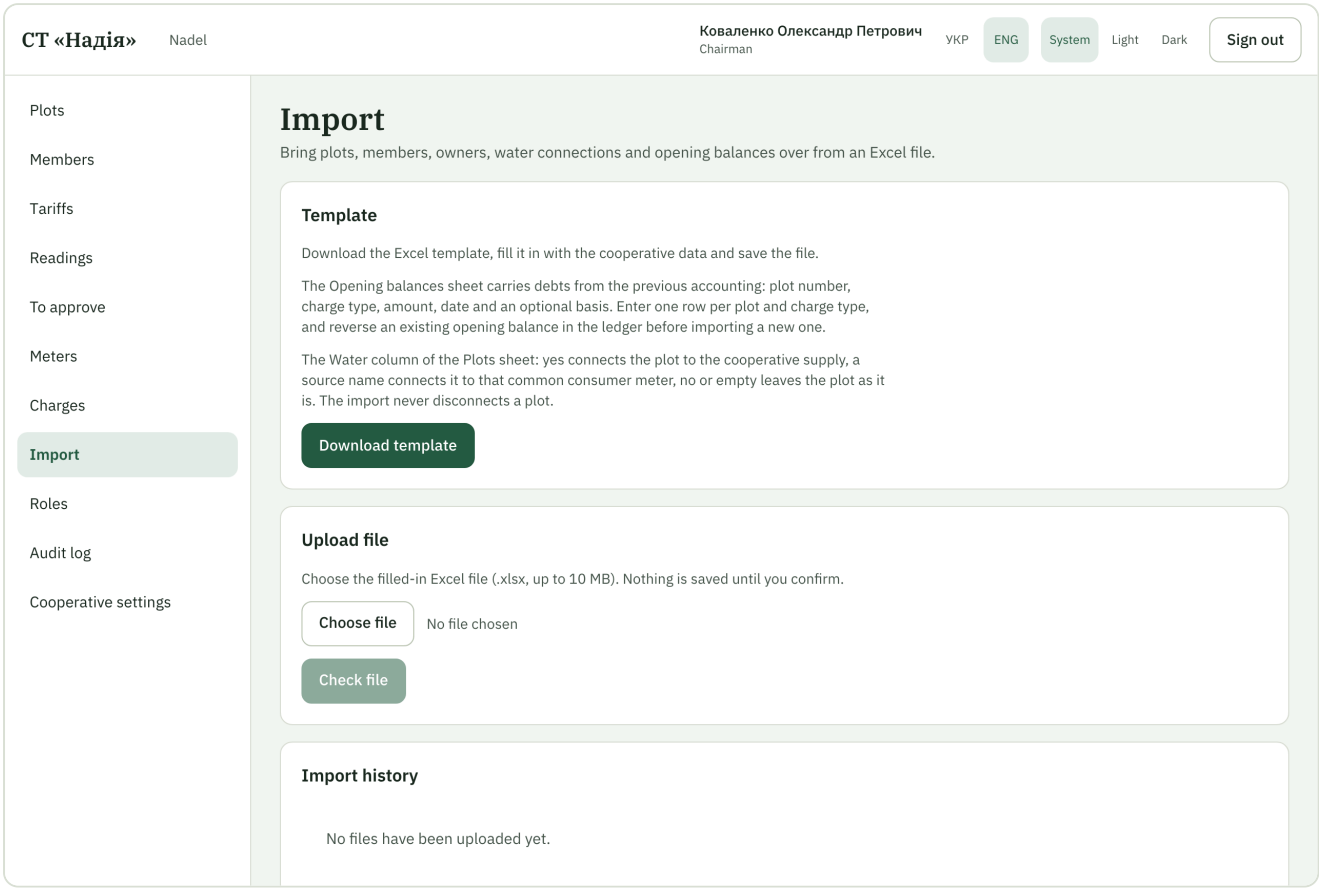
The meter tree is opened from here too — see chapter 9.

Charging settings

Below the colour choice: the red status threshold, how losses are charged, the split of common consumers and the water supply of plots. They are described in chapters 17–19.

7 Excel import

Instead of typing the cooperative in by hand, plots, members, owners and meters can be loaded from one Excel file.



The import page

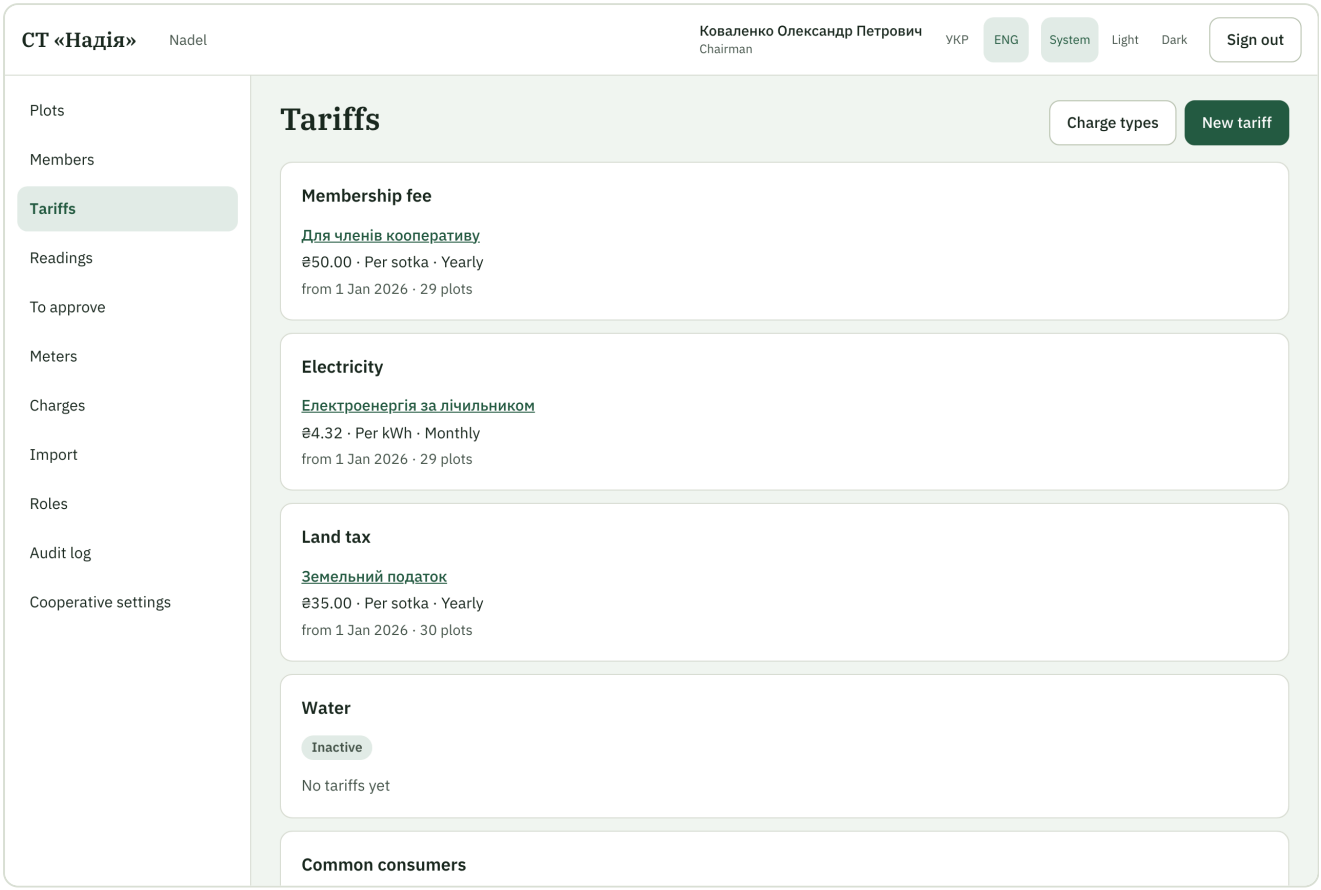
1. "Download template" gives a file in the interface language with the sheets Plots, Members, Ownership and Meters, each with a hint and an example row.
2. Fill in the sheets. Dates as Excel dates or `DD.MM.YYYY`, area in sotkas or m², shares as `1/2`.
3. Upload the file. The system checks everything **without saving** and shows how many records will be added, updated or left unchanged, and every problem with its sheet, row and column.
4. With no problems, confirm. Everything is saved together: the whole file or nothing.

- Importing the same file again duplicates nothing: plots are matched by number, members by email (or name and phone).
- An import cannot change an owner; use "Change owner" (chapter 5).
- Limits: `.xlsx` up to 10 MB, up to 2,000 plots and 3,000 members. A check is valid for 30 minutes.
- Sheet and column names are accepted in Ukrainian and English, even mixed.

Two additions: the **Opening balances** sheet (plot, charge type, amount, date, basis; one row per plot and charge type; amounts with at most two decimals) and the **Water** column of the Plots sheet ("yes" connects to the cooperative supply, any other text names a well; "no" or empty changes nothing). Importing the same file again books nothing twice.

8 Tariffs

Here the chairman describes what the cooperative charges for and how much. The charges themselves come later (chapter 17), but every plot card already shows what the plot will be charged.



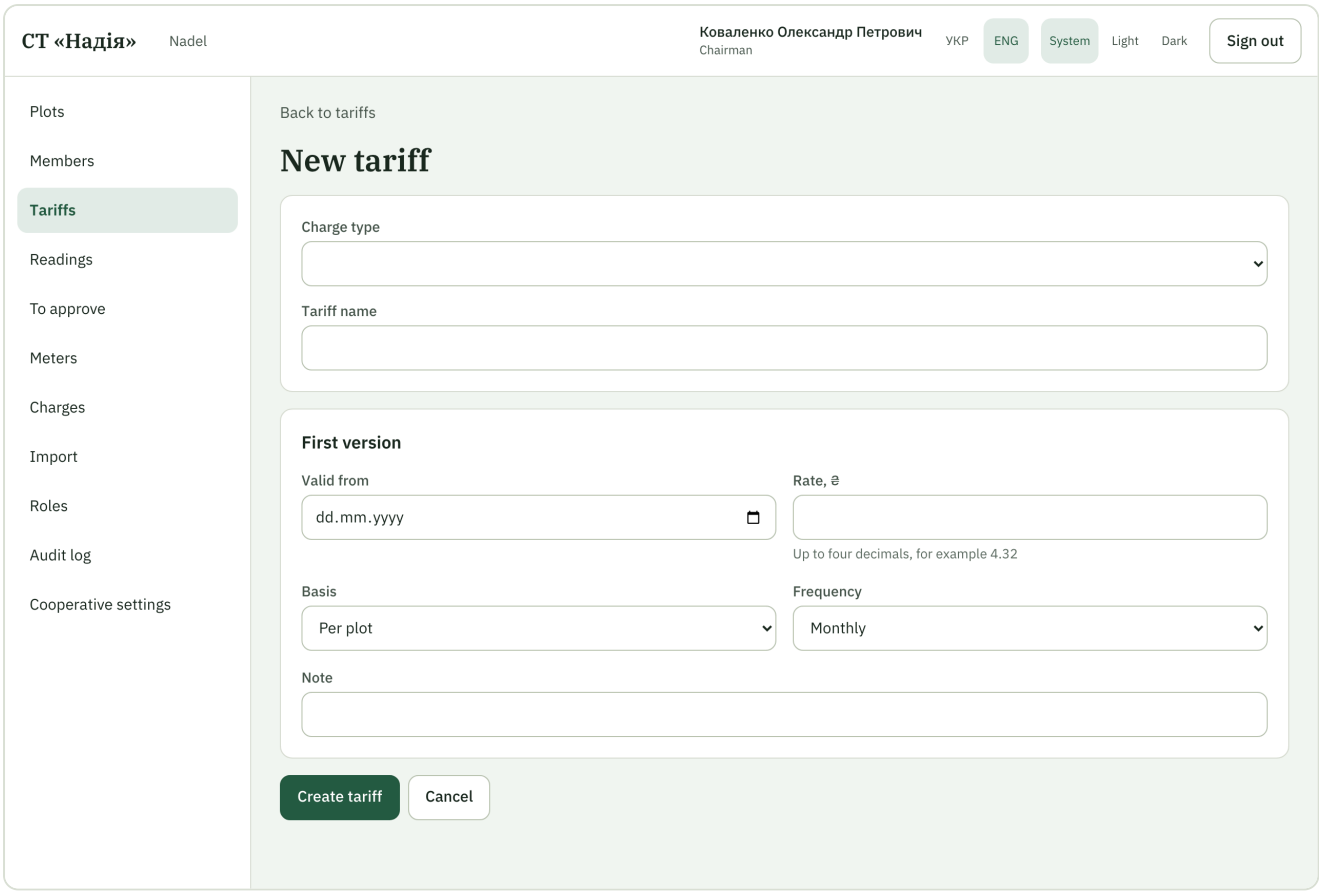
Tariffs by charge type

Charge types

The "Charge types" button. Every cooperative starts with membership fee, electricity, land tax and water (inactive until there are water meters). Add your own, such as "Road repair", rename, reorder, archive.

A new tariff

1. "New tariff" → choose the charge type and name it, for example "For members".
2. Fill in the first version: from which date it applies, the basis, the frequency and the rate.
3. Save, then on the tariff card choose the plots it applies to.



The new tariff form

Basis	Example
Per plot	300 ₴ per plot per quarter
Per sotka	Membership fee 50 ₴ per sotka per year
Per kWh	Electricity 4.32 ₴ per kWh, by meter readings
Per m³	Water, by water meter readings
Fixed one-time amount	Special-purpose fee 1,200 ₴ with a due date

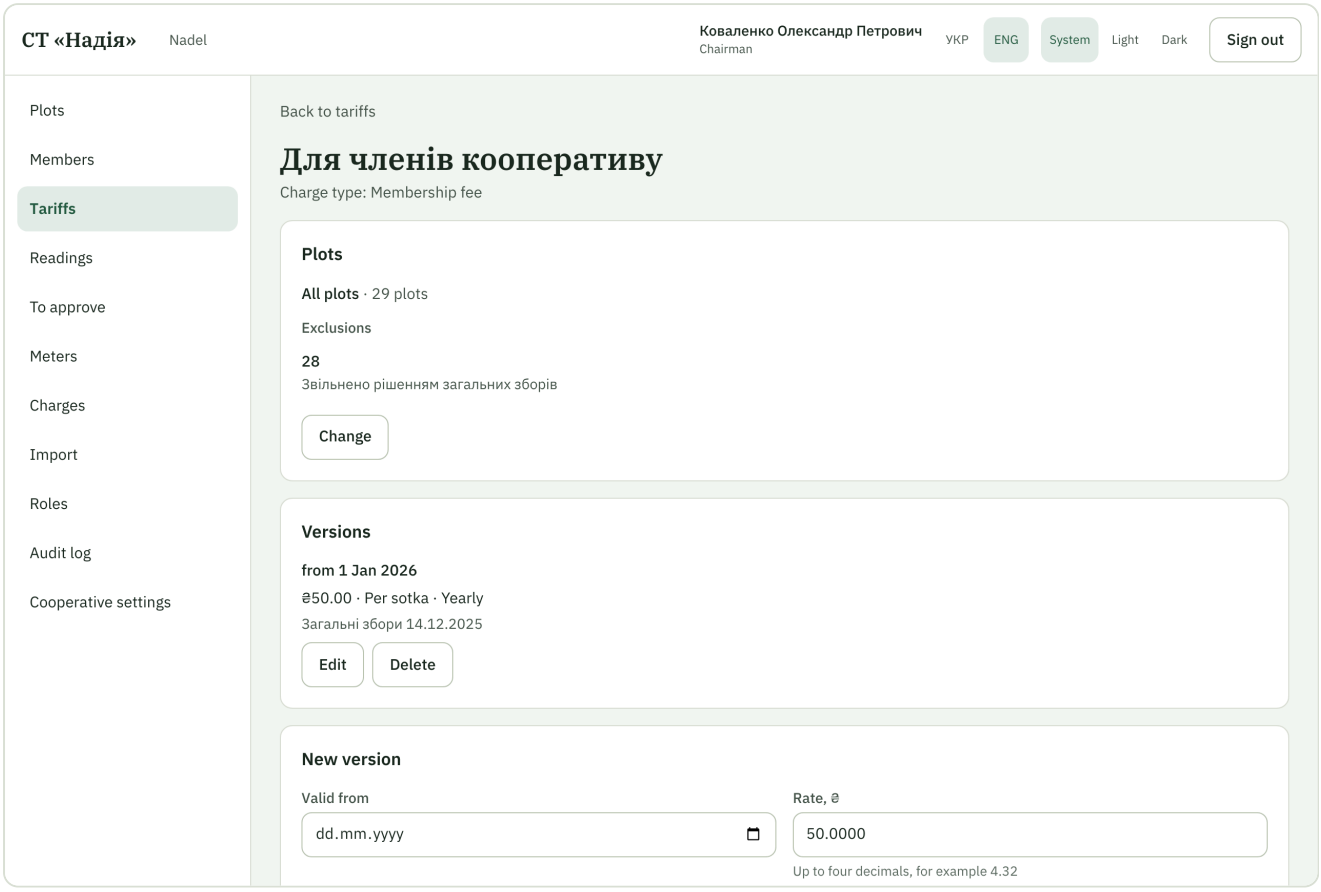
Frequency: monthly, quarterly, yearly or one-off. The cooperative bills monthly, so a yearly tariff is divided by 12 and a quarterly one by 3.

> **EXAMPLE**
>
> Membership fee 50 ₴ per sotka per year, a 6-sotka plot: 50 × 6 / 12 = **25.00 ₴ per month**.

A rate may have up to four decimals (for example 1.4532 ₴ per m² for land tax); amounts are rounded to kopecks only at the end. A rate is shown without trailing zeros: 50.00 ₴, not 50.0000 ₴.

Tariff versions

A tariff is changed not by editing but with a **new version** and the date it starts ("New version"). The version history is on the tariff card. A version starting in the past shows a warning that past charges will be recalculated.



A tariff card: versions and plots

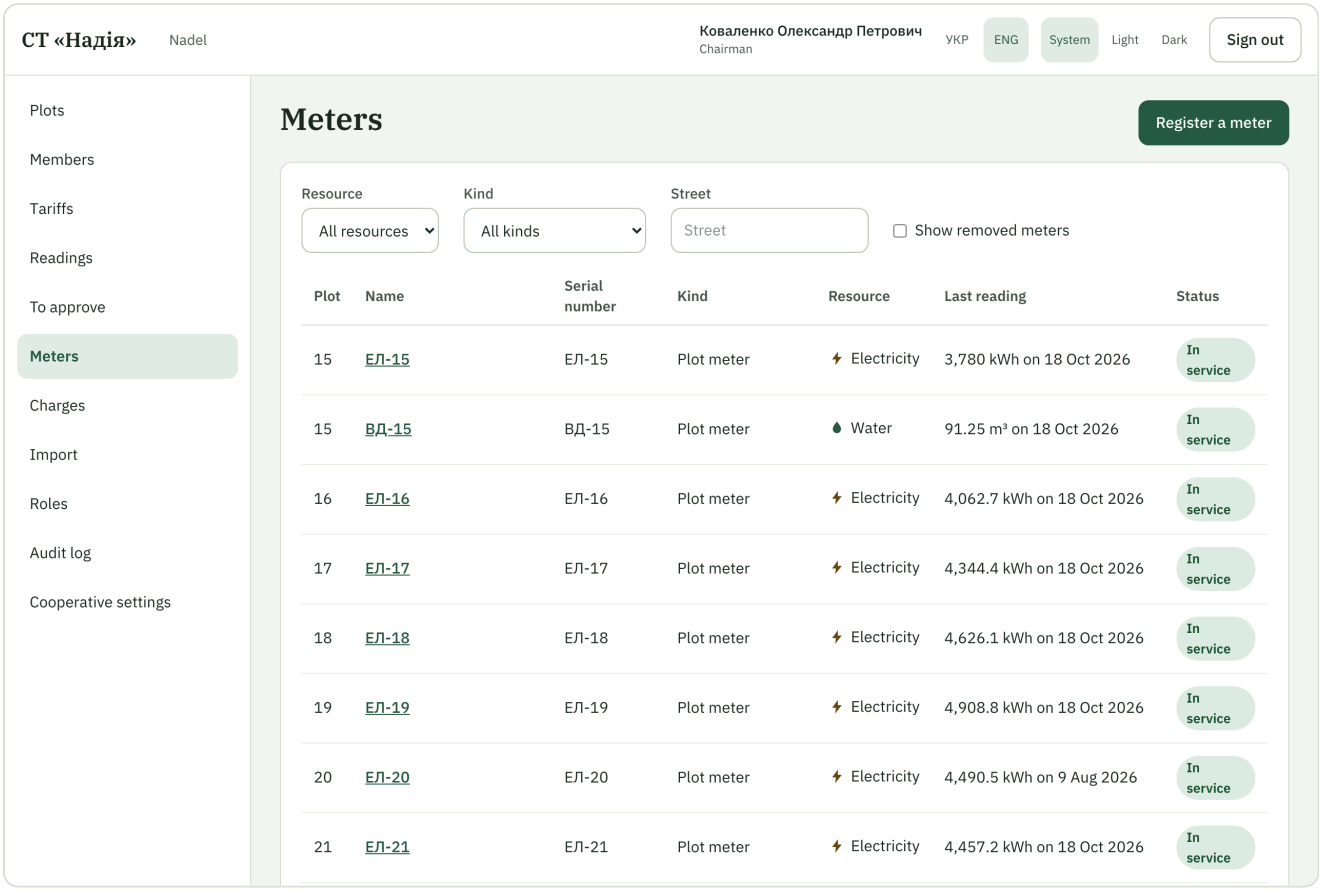
A mistake in a version can be fixed ("Edit") or the version deleted **while no charge uses it**. The only version of a tariff cannot be deleted; archive the tariff instead.

Which plots it applies to

- **All plots** (never common land), with **exclusions** if needed, each with a mandatory reason such as "exempted by the general meeting".
- **Selected plots**: an explicit list.
- Per-kWh and per-m³ tariffs apply only to plots with a meter of that resource. A plot without one shows "no meter".

9 Meters and the meter tree

"Meters" is the register of every meter in the cooperative. Each one has a resource icon: a **lightning bolt** for electricity, a **drop** for water.



The meter register

Kinds of meters

Plot meter	Belongs to a plot. A plot has at most one active meter per resource.
Substation	The meter at the cooperative's transformer: the root of the tree.
Quarter meter	The meter of a group of plots (a street or a block).
Common consumer	A well pump, street lighting, the office.

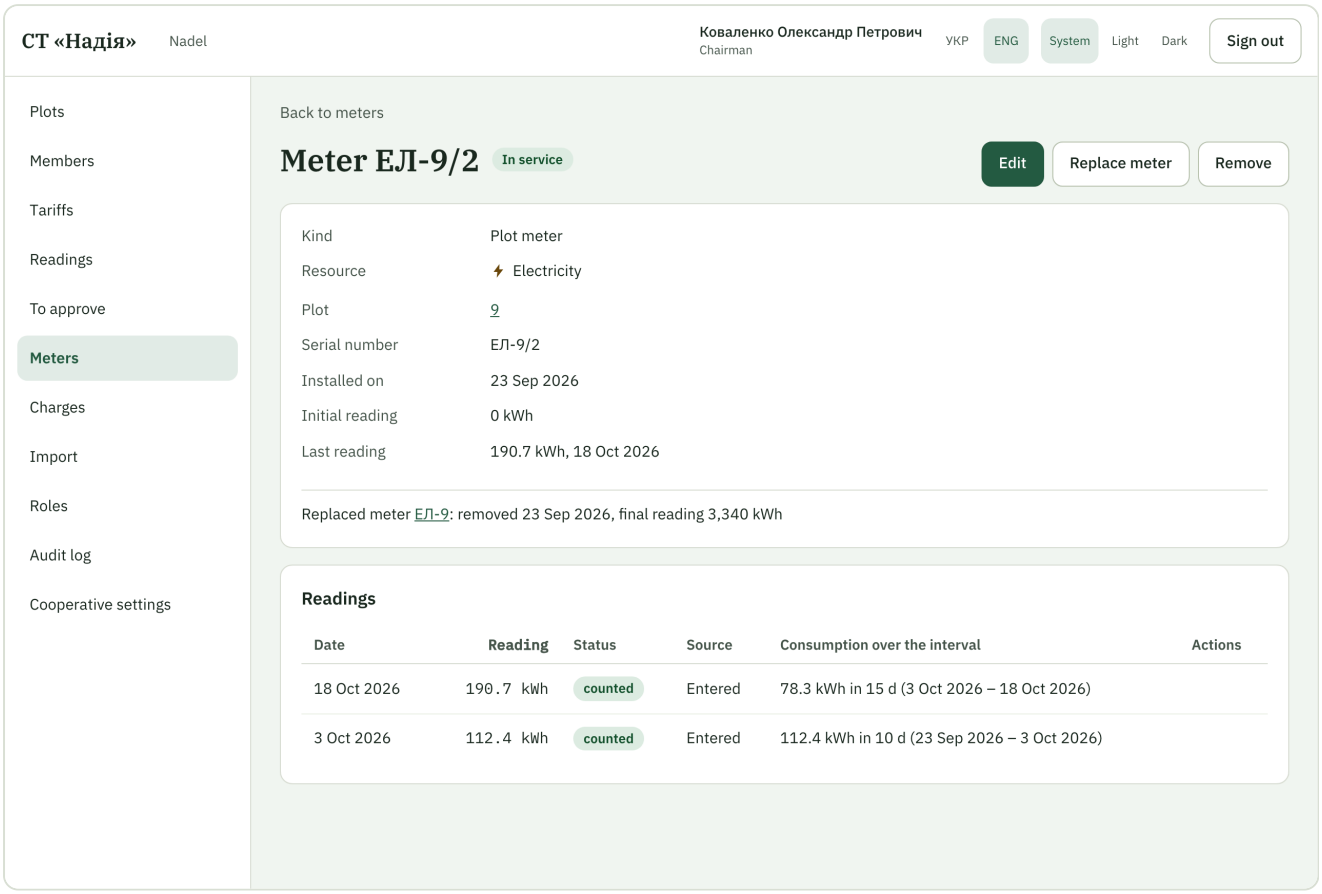
A cooperative may have any number of substations, quarter meters and wells — the system sets no limit.

Register, replace, remove

- **Register a meter**: kind, resource, serial number, plot (or a name for other kinds), installation date, initial reading.
- **Replace meter** is one action: the date, the final reading of the old meter, the serial number and initial reading of the new one. Both keep their history, and their cards link to each other ("Replaced

a previous meter…", "Replaced by the meter ЕЛ-9/2").

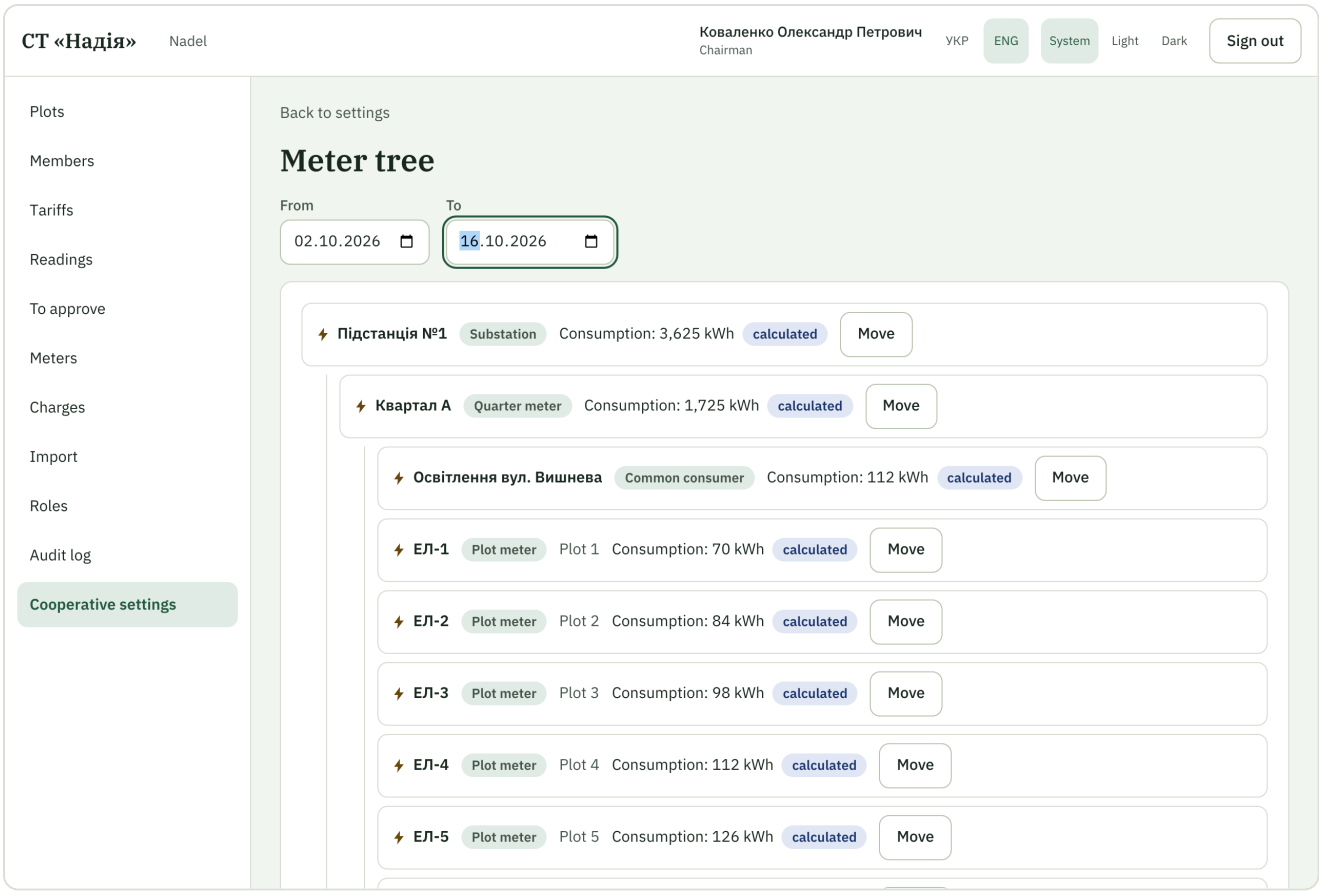
- **A counter wrapping past zero** (99 999 → 00 012) is a replacement of the meter by itself: "Fill in as a wrap-around" in the replace form.
- **Remove**: the meter stops working on the removal date; its readings are kept. Only a meter registered by mistake, without readings, can be deleted.



A meter card with its readings

The meter tree

"Cooperative settings" → "Meter tree". The chairman builds the tree: substations, below them quarter meters, common consumers and plot meters. "Move" puts a meter under another node **from a given date**; earlier periods use the old tree.



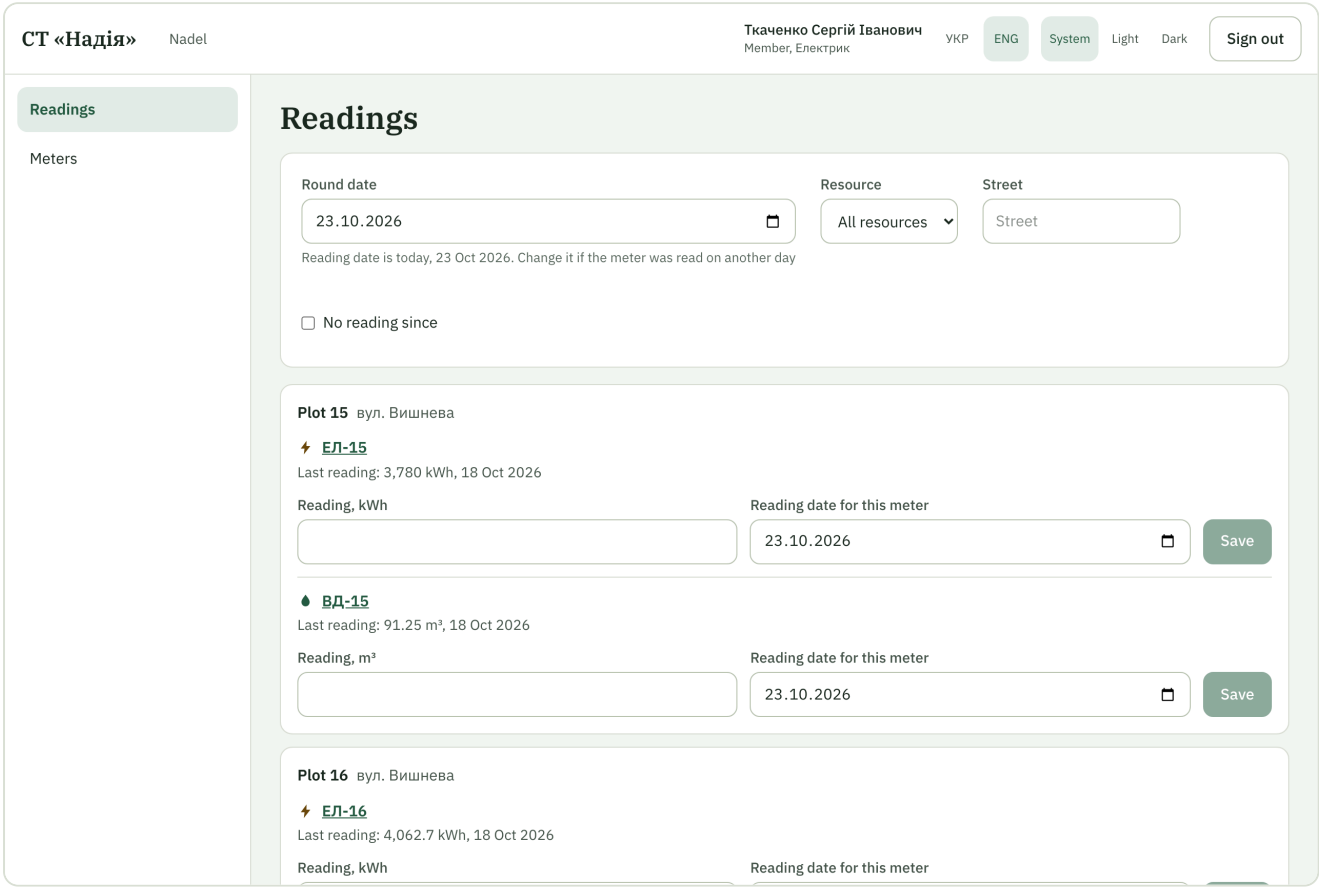
The meter tree with consumption for a range

Choose a range "from … to …" and every meter shows its consumption for it. With no reading on a range boundary, the system estimates the value in proportion to the days between the neighbouring readings and marks it "calculated". With no reading after the range end yet it says "no reading after …".

10 Readings: the round and approvals

Readings are not tied to months: the electrician may come after a month, after three or after a few days. **A reading is a meter's value on a date.** Consumption is counted between neighbouring readings.

The electrician's round

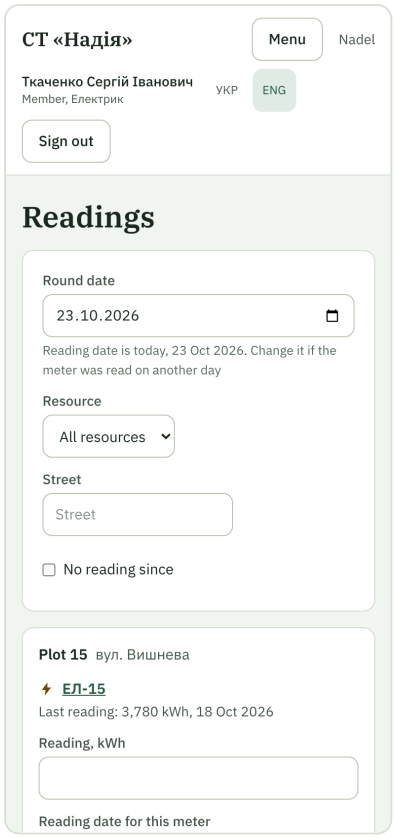


"Readings": one card per plot, a row per meter

1. Open "Readings". "Round date" already holds today's date with the hint "Reading date is today... Change it if the meter was read on another day". A single meter's date can be changed in its row.
2. Filter by resource (lightning / drop) if needed — the choice is remembered — and by street.
3. Each meter shows its last reading and date. Type the new reading and press "Save" in that row. The interval consumption appears at once.

- A reading lower than the previous one is refused: "The reading is lower than the previous one or higher than the next one".
- When consumption is much higher than usual, the reading is saved with the warning "Check the reading".
- "No reading since" lists meters that have not been read for a long time.

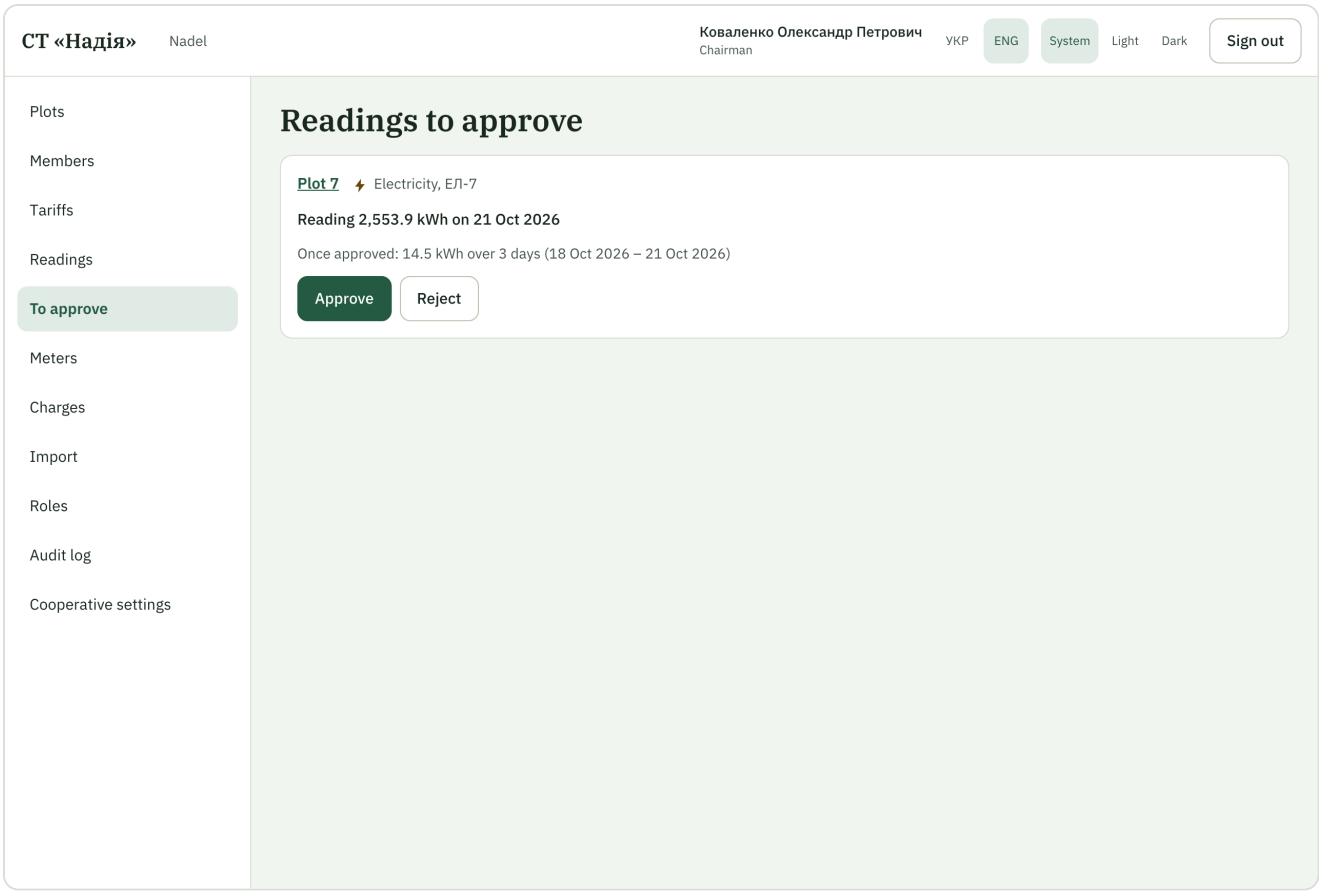
- A second reading for the same date replaces the first; the old one stays in the history marked "superseded".



The same screen on a phone

Readings from members

A member can submit the reading of their own meter from their account. It waits in "To approve": "Approve" or "Reject" with a reason the owner sees. A reading that is not approved is never used.



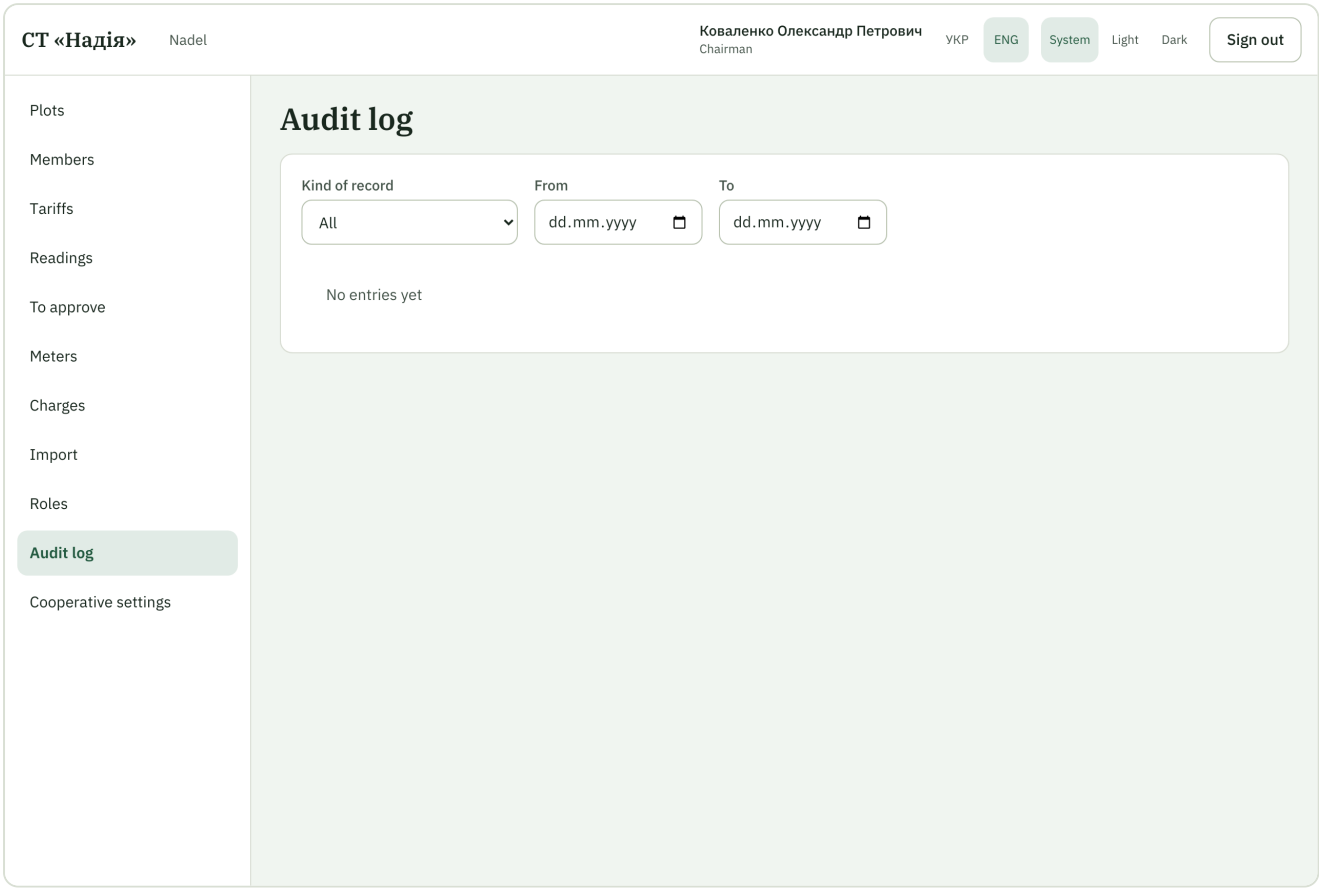
Readings to approve

Corrections

On the meter card every reading can be corrected or deleted **while no charge uses it**. Once charged, a correction will go through recalculation (chapter 20).

11 Audit log

Every board action is recorded: who, when, what changed — with old and new values. Filters by person, action, object and date.

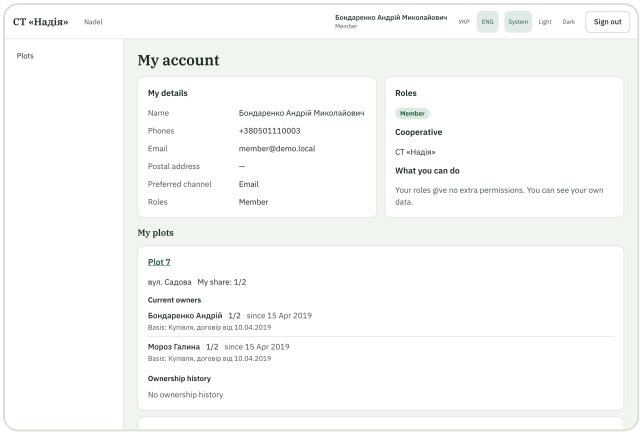


The audit log

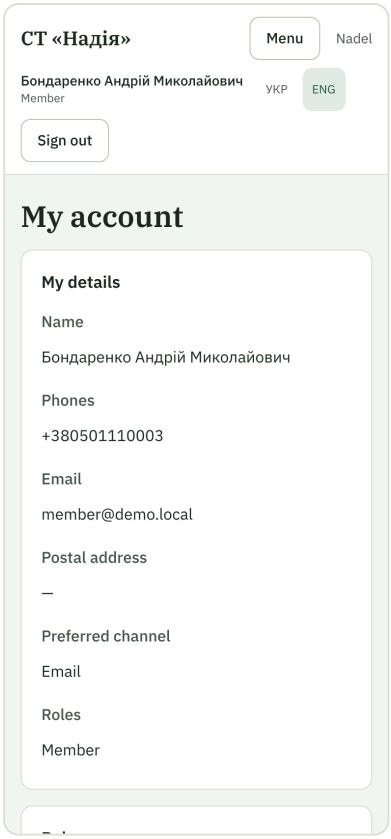
Log entries are never changed or deleted. Data loaded by demo scripts does not appear in the log — only actions of people in the system.

12 The member's account

After signing in a member sees "My account": own plots, what they will be charged by the tariffs, own meters with their readings and "Submit a reading".



My account

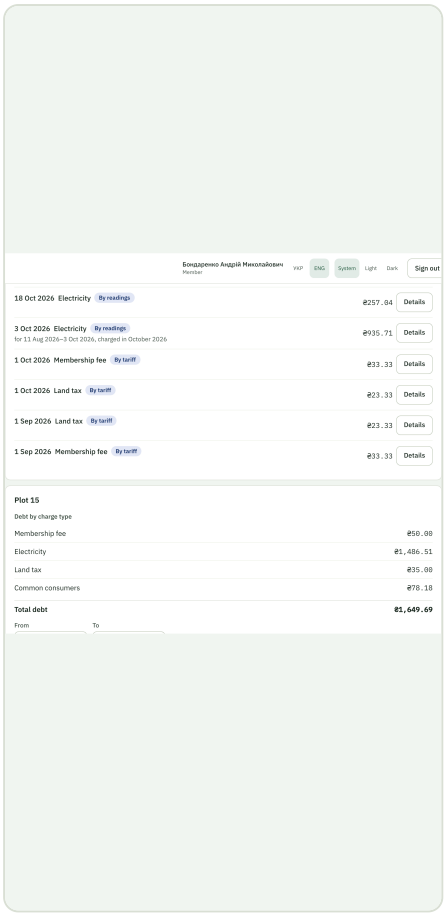


On a phone

A member never sees other plots' amounts, owners or personal data. Other plots' colours only when the chairman chose "All plots" (chapter 6).

My charges

For each of their plots the member sees the debt per charge type and every charge with its calculation, read-only. See chapter 17.



"My charges" in the member account

Part II

How the system calculates

The rules the system calculates amounts by, and why they are what they are — with worked examples. Chapters marked "planned" describe the next capabilities as they have already been decided.

13 Core principles

Only the server counts money	The web page and the app only show amounts, debts and statuses; they never calculate them. One formula, in one place.
Kopecks	Hryvnia with two decimals. Intermediate calculations are exact, never rounded. An amount is rounded to kopecks **once**, half up, only when a charge is produced.
A rate is not money	A tariff rate has up to four decimals (1.4532 ₴ per m²) and is never rounded. A fifth decimal is rejected, not cut.
Nothing is erased	Charges, owners and readings are never edited in place. A mistake is fixed by a new record (a reversal, a new version, a new reading); the old one stays in the history.
Audit log	Every board action is recorded with old and new values.
Cooperatives apart	Each cooperative's data is separated in the database itself: another cooperative cannot see it even if the code has a bug.
One language	The interface shows only the selected language, never a mix.
Minimal personal data	No tax numbers, passports or dates of birth; personal data never goes to the server logs.

> **WHY ROUND ONCE**
>
> A rate of 0.05 ₴ for 15 + 15 days of a 30-day month is exactly 0.05 ₴. Rounding each half separately gives 0.03 + 0.03 = 0.06 ₴, a kopeck more. Over thousands of plots and months such kopecks add up.

14 Tariffs and the expected amount

A charge type (membership fee, electricity, land tax, water, special-purpose fees) has one or more tariffs. A tariff has **versions**: each applies from its date until the day before the next version. Two versions never start on the same day.

```
units           per plot → 1 · per sotka → area m² / 100
period share    monthly 1 · quarterly 1/3 · yearly 1/12
month amount    rate × units × period share × (days the version applies in the month) /
(days in the month)
one-time fee    the whole rate in the month of its due date, 0.00 in other months
```

Worked examples

Case	Given	Result
Membership fee	50 ₴/sotka/year, plot 600 m²	50 × 6 / 12 = 25.00
Odd area	50 ₴/sotka/year, 612 m²	50 × 6.12 / 12 = 25.50
Rounded at the end	33.3333 ₴/sotka/year, 700 m²	19.444425 → 19.44
Per plot, quarterly	300 ₴ per quarter	100.00 per month
Version change mid-month	50 ₴ until the 15th, 60 ₴ from the 16th, 600 m², 30 days	12.50 + 15.00 = 27.50
Special-purpose fee	1,200 ₴, due 01.05.2026	1,200.00 in May, 0.00 otherwise
Land tax	35 ₴/sotka/year, 600 m²	17.50 per month

Which plots

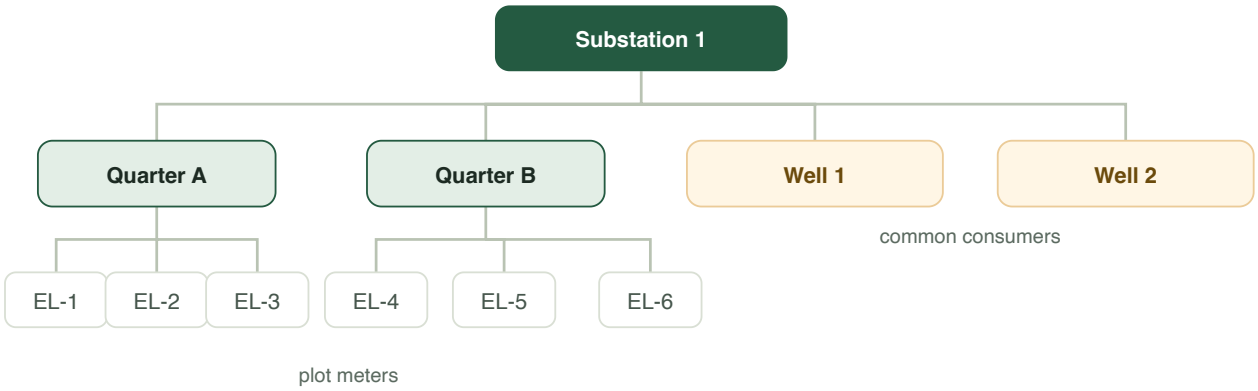
- "All plots" minus exclusions (each with a reason), or "Selected plots". Never archived plots or common land.
- Per-kWh and per-m³ tariffs only for plots with an active meter of that resource. **No meter, no charge** (confirmed by the cooperative's chairman).

Correcting a version

A version can be corrected or deleted **while no charge uses it**. After that a correction goes through recalculation (chapter 20): charges already issued to people never change silently.

15 Meters and the tree

A meter measures one resource: electricity (kWh) or water (m³). Meter values are not money: up to three decimals.



The meter tree: substation → quarters → plots; common consumers are nodes of their own. There may be any number of substations, quarters and wells.

- The chairman builds the tree; nothing about it is hard-coded. Moving a meter under another node takes effect from a date; the tree's history is kept, so a past range is calculated with the tree as it was then.
- Any number of substations, quarter meters and common consumers (wells included); every quarter meter and every well has its own list of plots.
- A plot has at most one active meter per resource.

Replacing a meter

A replacement is one action: the final reading of the old meter, the serial number and the initial reading of the new one. Consumption of a period with a replacement is the sum over both meters.

> EXAMPLE
>
> Old meter: 8,140 at the start, removed at 8,205. New one: installed at 3, reads 69 at the end. Consumption: (8,205 − 8,140) + (69 − 3) = 65 + 66 = **131 kWh**.

A counter wrapping past zero (99,999 → 00,012) is "a replacement of the meter by itself": same serial number, final 99,999, initial 0. The same rule then applies.

16 Readings and consumption

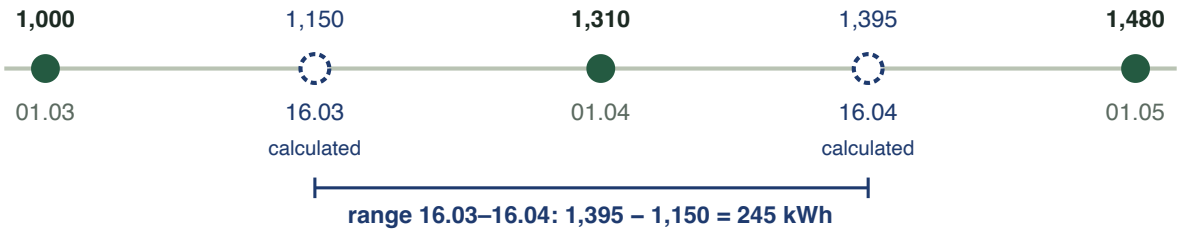
A reading is a meter's value on a date, not "the reading for a month". The electrician comes when they can: after a few days or after a few months. The person enters the date; it defaults to today, with an unobtrusive hint.

Reading statuses

Status	Meaning	Counted
Approved	Entered by the electrician or the board, imported, or a member's reading approved	yes
Pending	Submitted by a member, waits for approval	no
Rejected	With a reason the owner sees	no
Superseded	Replaced by a new reading for the same date or a correction; kept in the history	no

Intervals and the value on a date

- **Interval consumption** is the difference between a reading and the previous reading of the same meter.
- **The value on any date**: the real reading if there is one on that date; otherwise a linear interpolation by days between the neighbouring readings, marked "calculated"; with no reading after the date yet, unknown.
- **Consumption for a range "from … to …"** = the value at the end − the value at the start.



Real readings (green) and values calculated at the range boundaries (dashed)

> **WHERE 1,150 COMES FROM**
>
> Between 01.03 (1,000) and 01.04 (1,310) there are 31 days and 310 kWh. 16.03 is day 15: 1,000 + 310 × 15 / 31 = 1,150. Calculated values are always marked so that nobody takes them for a real reading.

Checks

- A reading lower than the previous one (or higher than the next one, when entered back-dated) is refused. A lower value is valid only as the initial reading of a new meter.
- A date in the future or outside the meter's working period is refused.
- Daily consumption three times the average of the last 180 days: the reading is accepted with the warning "Check the reading".
- A reading can be corrected or deleted while no charge uses it; after that, through recalculation.

17 Charges and closing months works

A charge is an entry in the plot's account: charge type, amount, month, the days it pays for (service period) and how the amount was obtained (rates, units, kWh, the readings and tariff versions used).

Where charges come from

Source	How it arises
Recurring tariffs	The board presses "Charge the month": preview → "Post". A rerun adds only what is missing, never duplicates.
Electricity	Every approved reading at once gives a charge for the interval since the previous reading.
Network losses	Depending on the chairman's setting: a percentage on every electricity charge, or a board run for a range (chapter 18).
Common consumers	A separate board run for a range, a separate line in the account (chapter 18).
Opening balance	A plot's debt on the day the cooperative starts in the system — by hand or with an Excel sheet.

Electricity by reading

The days of an interval run from the day after the previous reading to the day of the new one, both included. When the tariff changes inside the interval, the kWh are split by days, each part times its rate, and the total is rounded once.

> EXAMPLES
>
> Readings 112.4 (22.09) and 205.7 (04.10): 93.3 kWh × 4.32 = **403.06 ₴** for 23.09–04.10 (12 days).
>
> Readings 1,000 (01.10) and 1,300 (31.10), tariff 4.32 until 16.10 and 5.00 from 17.10: 15 days each, 150 × 4.32 + 150 × 5.00 = 648.00 + 750.00 = **1,398.00 ₴**.
>
> A meter replacement gives two charges: 65 kWh = 280.80 and 66 kWh = 285.12.

Months and closing

- A month is open or closed. Months are closed only in order; a closed month never reopens.
- Before closing the board sees a check list: is the month posted, what was not charged, which meters have not been read for long. None of it blocks closing — a person decides with everything in view.

- A charge for an already closed month (a late reading, say) goes to the earliest open month, marked "for September, charged in October".

Reversals and the "not charged" queue

- Charges are never changed or deleted — the database itself enforces it. A wrong charge is cancelled by a **reversal** (the same amount with a minus and a reason). A whole losses or common-consumer run can be reversed together.
- What the system could not charge goes to the "not charged" queue with the reason: no tariff for some days; two tariffs of one type on a plot on the same day; no node readings. Once the cause is fixed, "Retry".

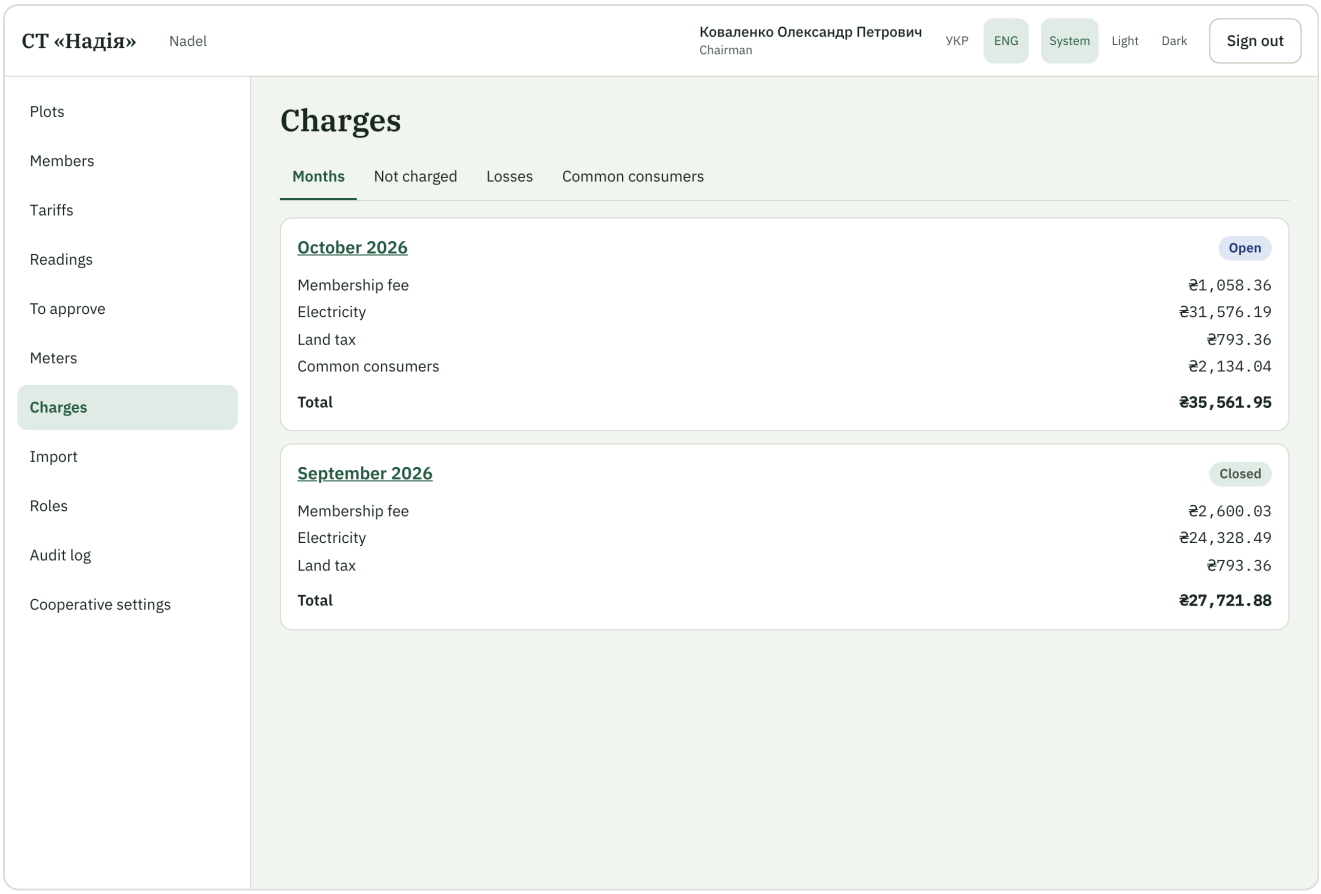
Debt

```
debt per charge type = Σ charges of that type (reversals negative) − Σ allocated
payments
prepayment           = Σ payments − Σ allocated payments
```

Payments arrive in the next phase. Until then a plot with charges shows a debt — that is correct.

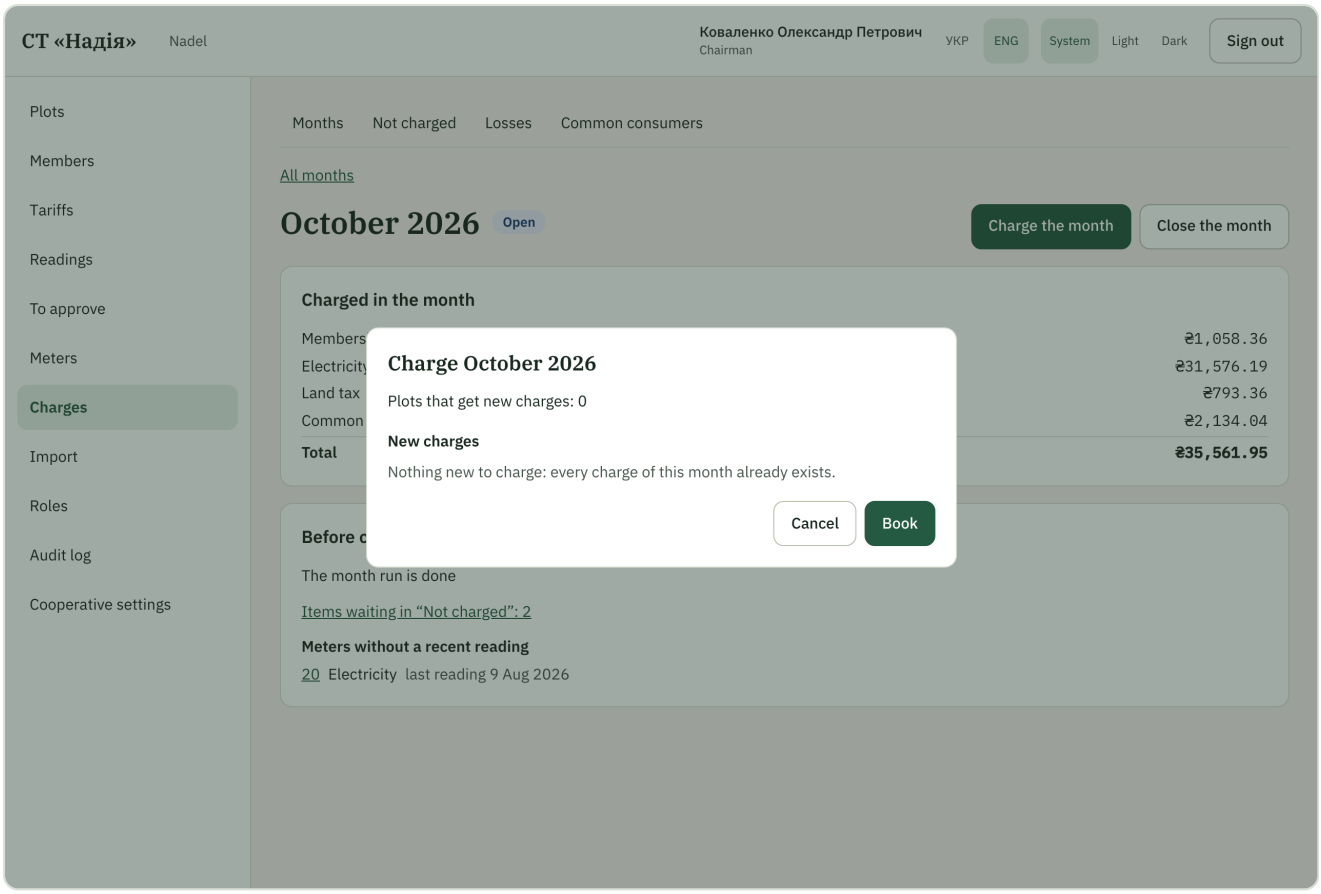
On the screen: "Charges"

The "Charges" menu item (permission "View charges") has four tabs: Months, Not charged, Losses and Common consumers.



"Charges": the months, newest first, with the total per charge type

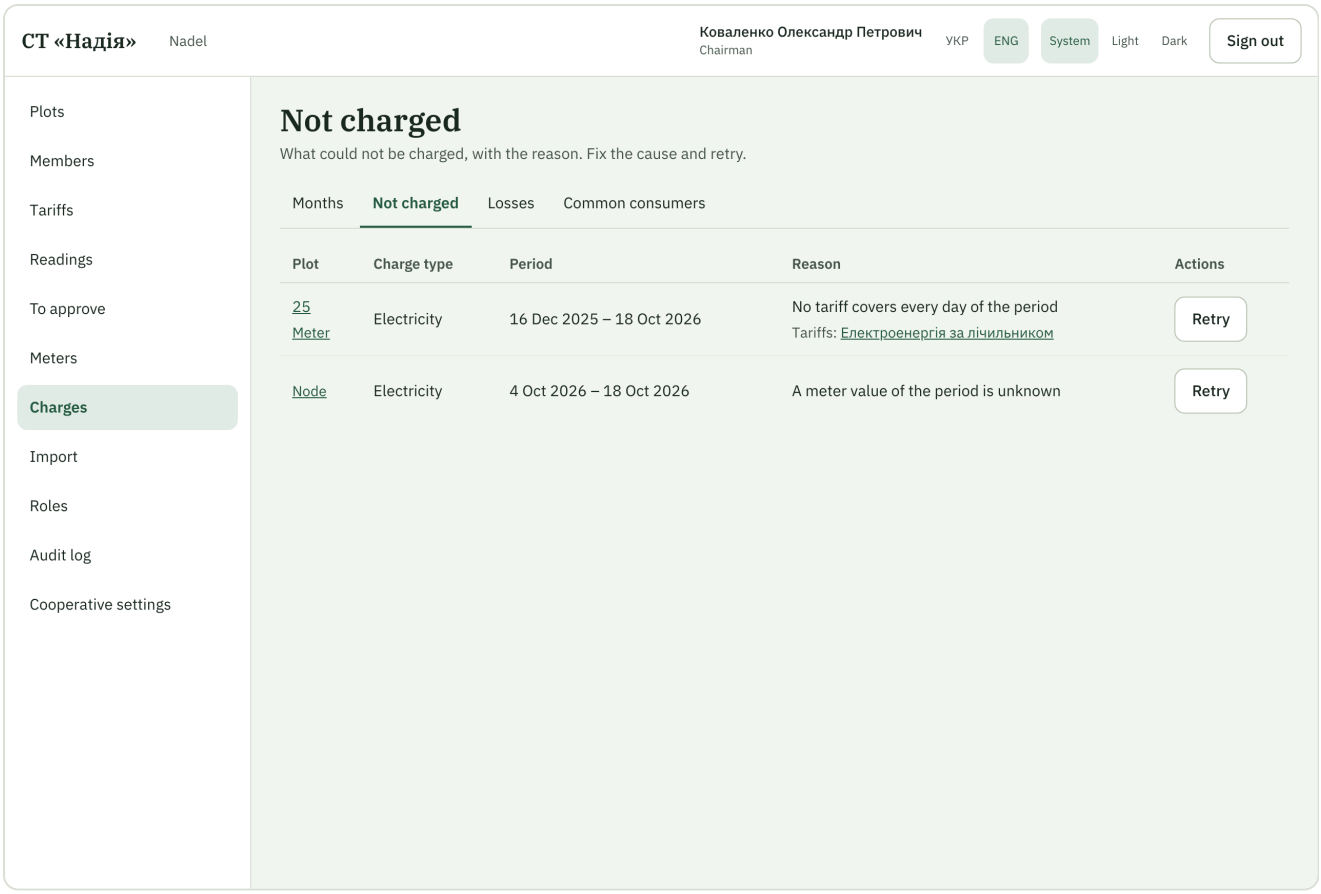
1. Open a month. The page shows what was charged per charge type and the check list ("Before closing"): is the month run done, how many items wait in "Not charged", which meters have no reading for more than 60 days.
2. "Charge the month" (permission "Run charges") first shows a **preview**: the number of plots, the new charges per charge type and the items that cannot be charged. "Book" books exactly what was shown. If anything changed meanwhile the system asks to preview again.
3. A rerun creates only what is missing: a plot or a tariff added later gets its charge, nothing is duplicated. A month after the current one cannot be charged.



The preview of the month run: nothing new, because October was already run

"Not charged"

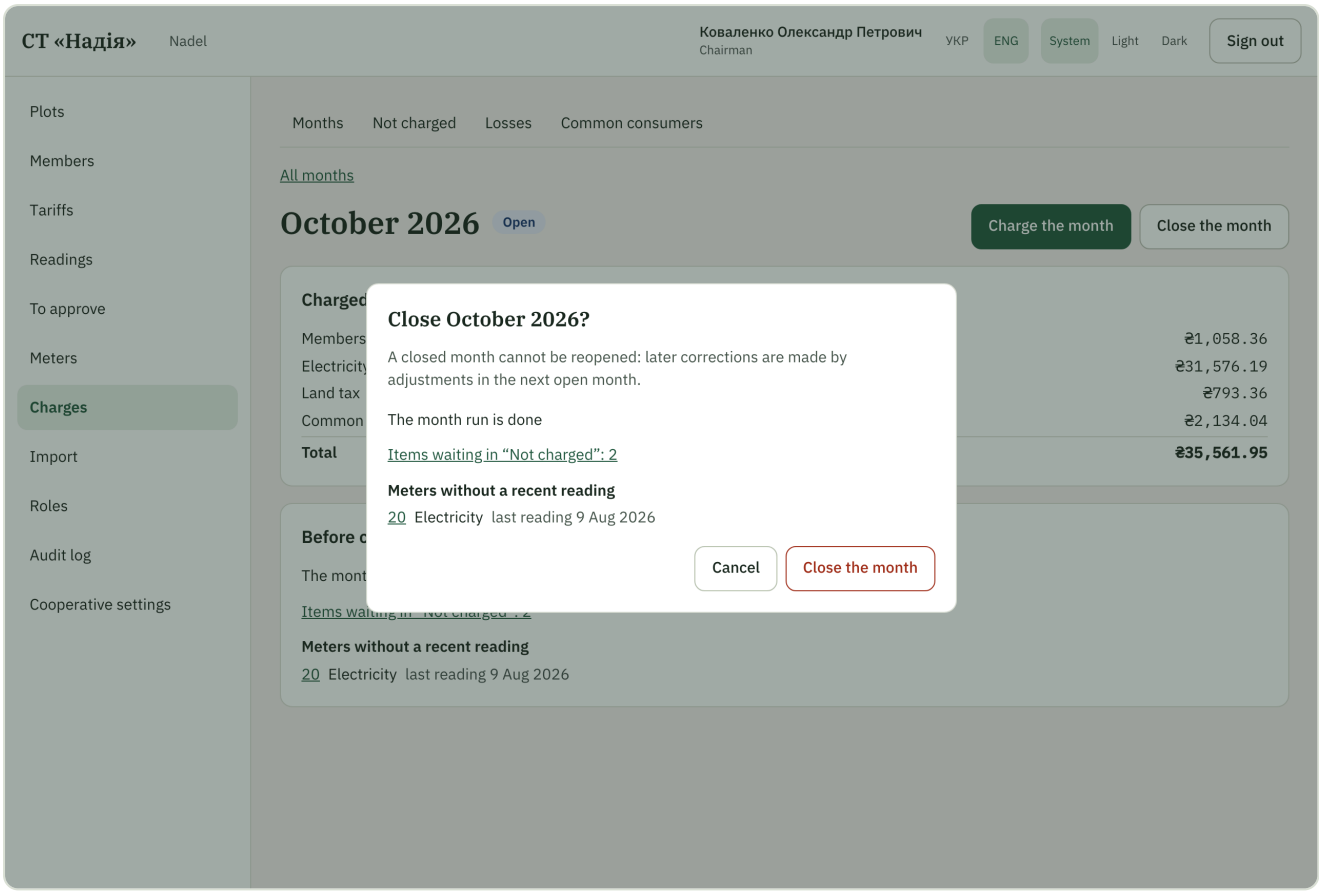
One list of everything the system could not charge, with the reason and a link to fix it: no tariff for every day, two tariffs on the same day, a plot outside every tariff, an unknown node or consumer value. "Retry" tries one item again; the next month run retries the tariff and reading items. An item disappears when the retry succeeds. Items never block closing a month.



"Not charged": the reason and "Retry" for each item

Closing a month

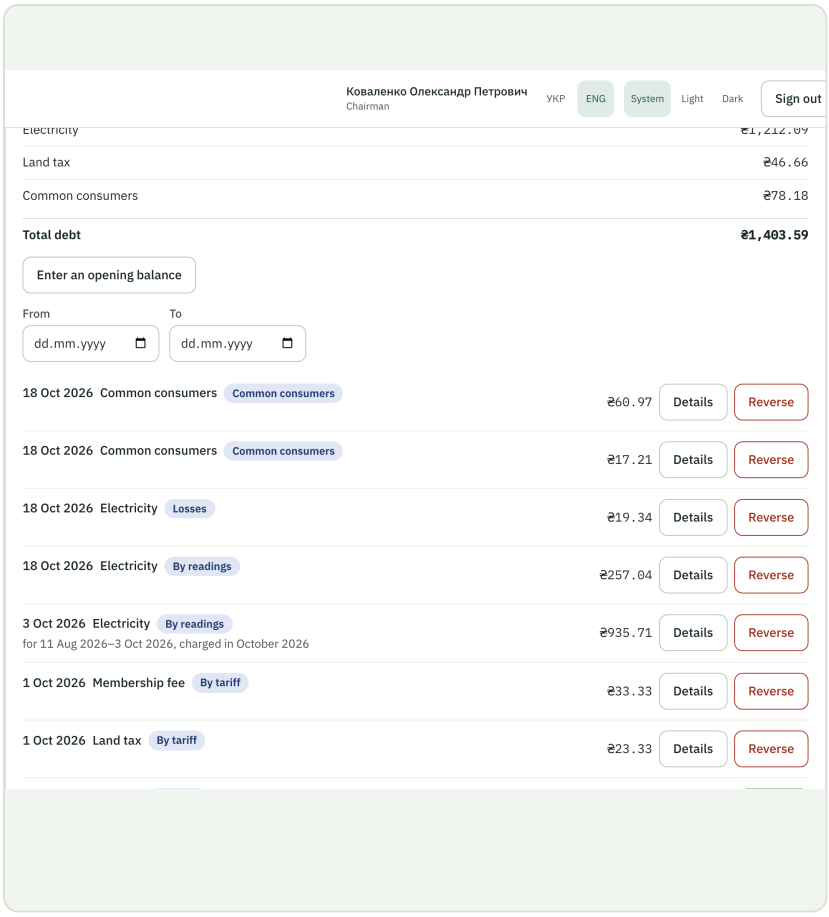
"Close the month" (permission "Close periods") shows the check list once more and asks to confirm. Only the earliest open month can be closed ("Close the earlier months first" otherwise). A closed month never changes; late entries go to the earliest open month.



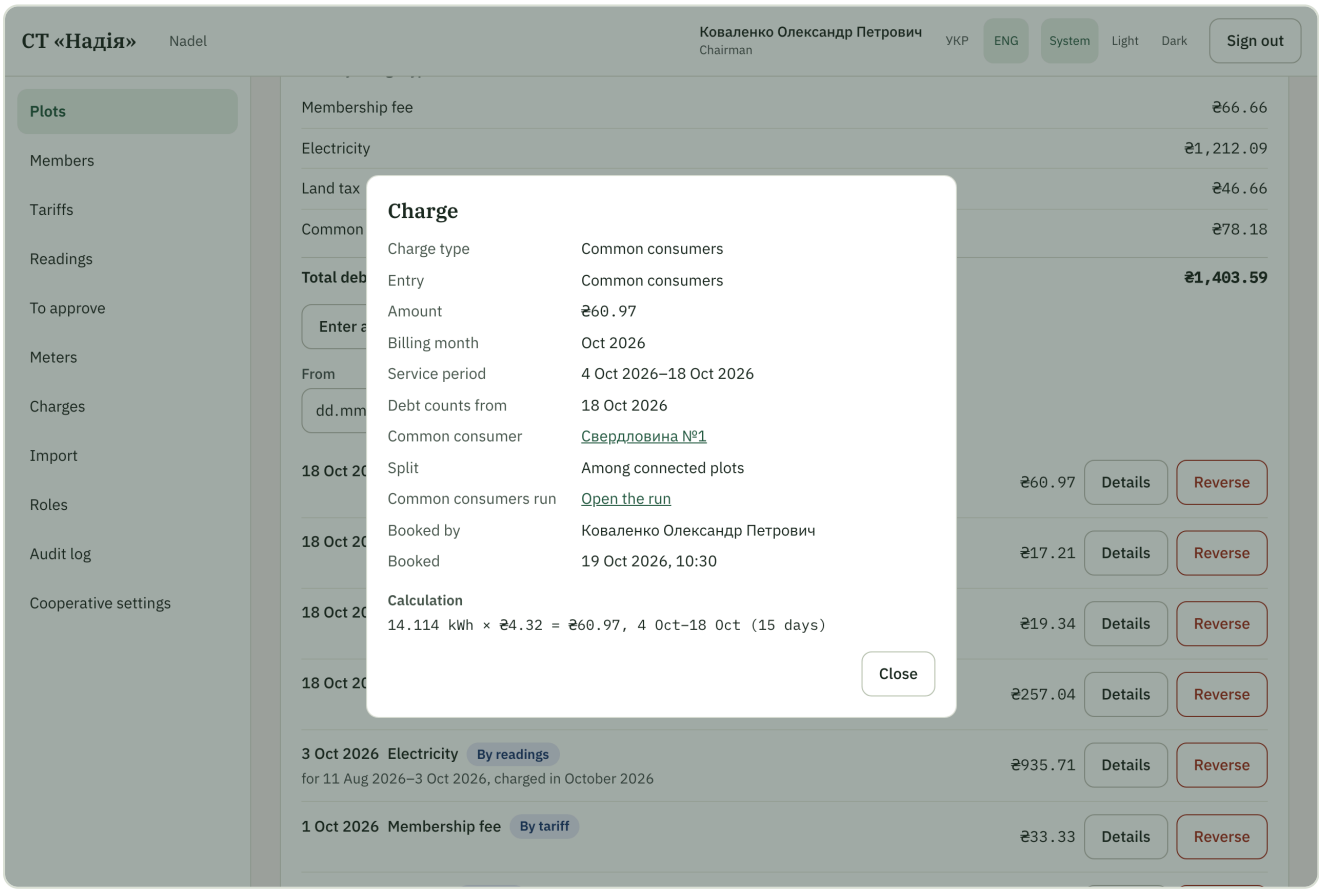
Closing: the check list stays in view, nothing blocks it

The plot account, reversals and opening balances

On the plot card the "Account" panel shows the debt per charge type and the charges, newest first, with a date filter. "Details" opens the calculation of a charge: kWh, rate, readings, days, the losses percentage. A charge booked later than the days it pays for reads "for 13.09–28.09, charged in October 2026".



The plot account



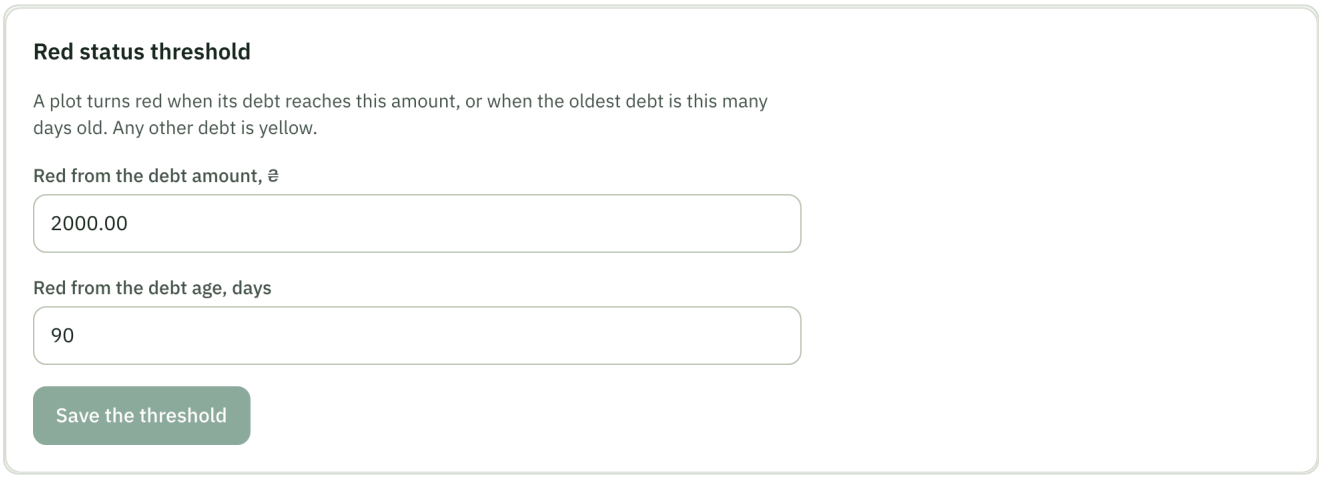
The calculation of a charge

- **Reverse** (permission "Reverse charges") asks for a reason and creates an entry for the same amount with a minus. A charge is reversed once; a reversal cannot be reversed; a charge in a closed month is corrected by an adjustment (a later phase), the button is not offered.
- **Enter an opening balance** (permission "Run charges"): charge type, amount, "debt counts from" (not later than today) and an optional reason. Many plots at once: the "Opening balances" sheet of the Excel import (chapter 7). One opening balance per plot and charge type; to change it, reverse it first.
- Reversing a whole losses or common consumers run is done on its page (below).

Plot colours on the map and in the list

Colour	When
Red	The debt reaches the "red from amount" or the oldest debt is at least "red from days" old
Yellow	Any other debt
Green	No debt
Grey	No owner, or common land

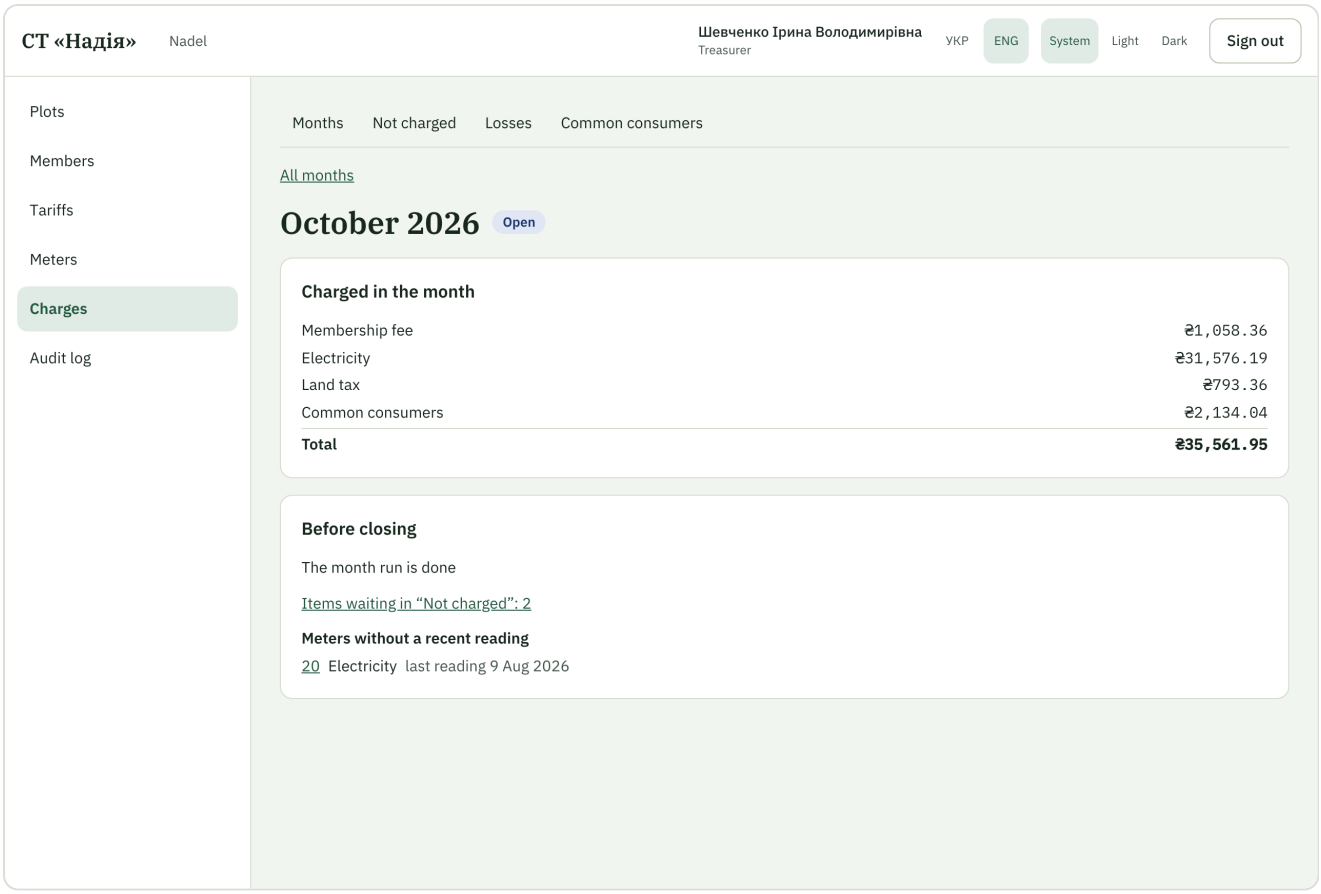
The thresholds are set in "Cooperative settings" (defaults 1,000.00 ₴ and 90 days; the demo cooperative uses 2,000.00 ₴ and 90 days).



The red status threshold in the settings

The treasurer and the member

The treasurer sees months, accounts and the lists but has no buttons. A member sees only the account of their own plots under "My charges" (chapter 12), read-only, with the calculation of every charge; other plots' amounts are never shown.



What the treasurer sees: no actions

18 Network losses and common consumers

works

Losses: two modes, the chairman chooses

- **Fixed percentage**: every electricity charge gets, for example, 5 % kWh on top. 65 kWh → losses 3.25 kWh = 14.04 ₴.
- **Proportional by node**: the board chooses a range and runs the calculation. For every node, bottom-up: losses = node consumption − the sum of its children's consumption. The losses are shared among the plots under the node in proportion to their consumption.

> EXAMPLE: A QUARTER AND THE SUBSTATION
>
> Quarter A: 1,100 kWh, plots P1 = 600 and P2 = 400 below it → losses 100: P1 60 kWh (259.20 ₴), P2 40 (172.80 ₴).
>
> Quarter B: 1,050, plot P3 = 1,000 → losses 50 → P3.
>
> Substation: 2,300; children A 1,100 + B 1,050 + well 50 = 2,200 → losses 100 shared 600 / 400 / 1,000 of 2,000: P1 30, P2 20, P3 50 kWh.

- **Negative losses** (the plots together "used" more than the node shows: a wrong reading, different reading dates, a faulty meter): nothing is charged, the board sees a warning with the numbers.
- **An unknown value** (no reading of the node or a child after the range end): that node and the nodes above it are skipped with a warning; the rest of the tree is charged.
- Two runs cannot cover the same days of the same node.

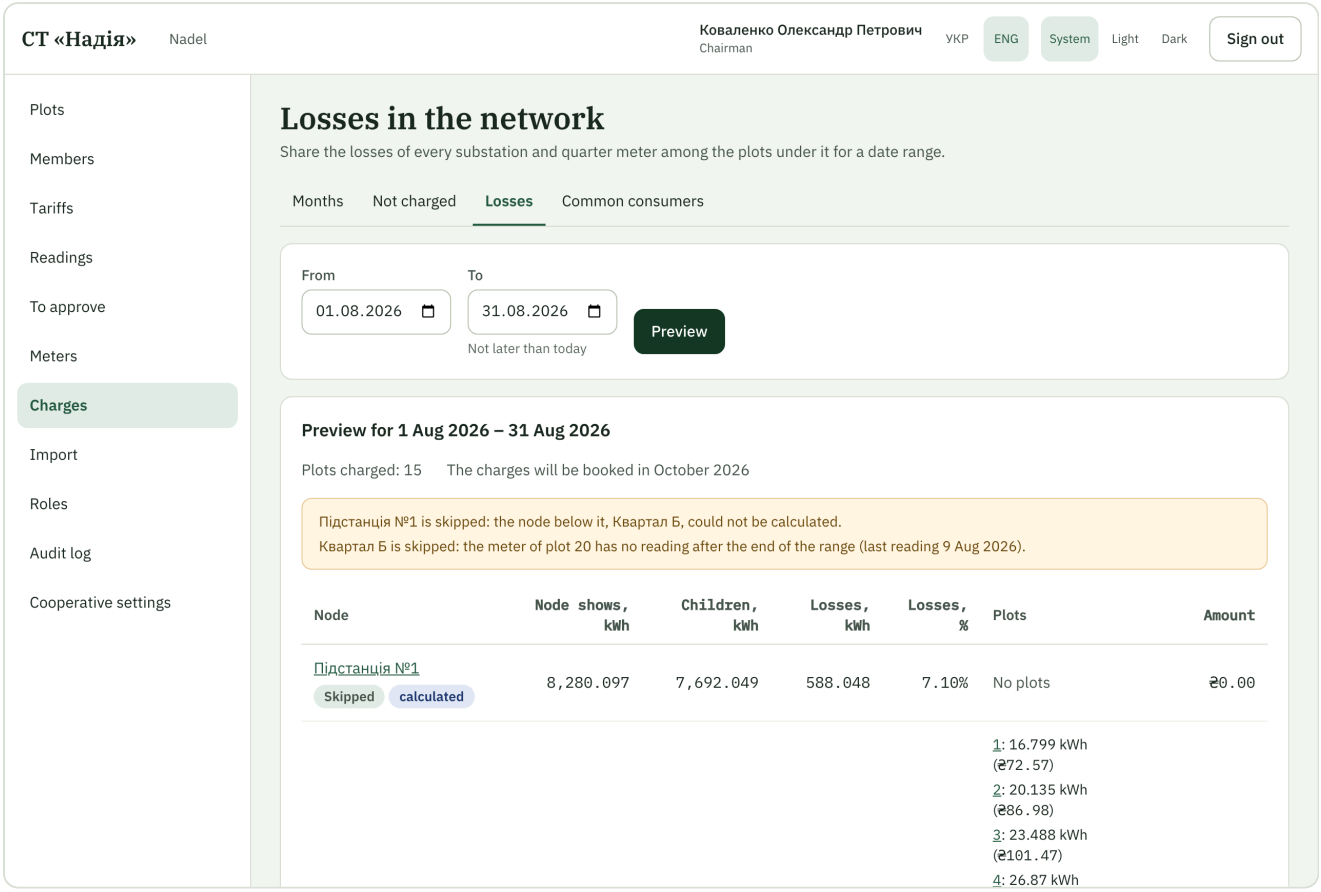
Common consumers — a separate line

Well pumps, street lighting, the office. Their consumption is a separate charge type "Common consumers" with its own per-kWh tariff and its own board run. The split is chosen per consumer:

Split	Who pays	Example
Connected plots	Plots connected to that well, for their connected days	Pump 60 kWh, three connected plots: 20 kWh each = 86.40 ₴; an unconnected plot nothing
Equally	Every active plot, with or without a meter	Lighting
Service users	The plots listed in the settings	Lighting 30 kWh, 3 plots: 10 each = 43.20 ₴
Not charged	Nobody — covered by the membership fee	Shown in reports

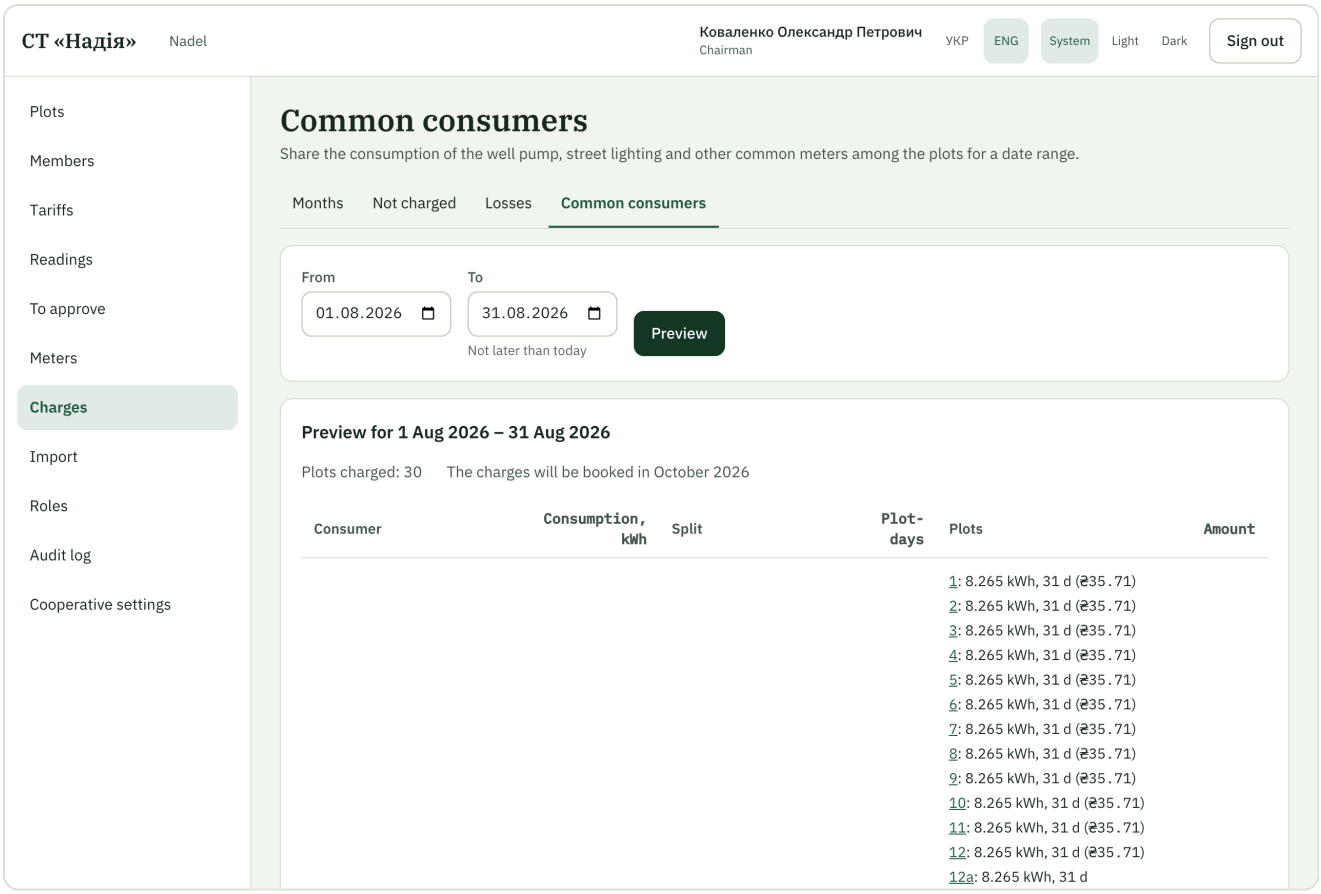
On the screen: "Losses" and "Common consumers"

Both tabs of "Charges" work the same way: pick the range "From" and "To" (not later than today), press "Preview", read the table and warnings, then "Book" (permission "Run charges"). The booking is written in the month of the "To" day, or in the earliest open month after it. The history of past runs is below, with "Details" and "Reverse the run" (permission "Reverse charges", a reason is required).

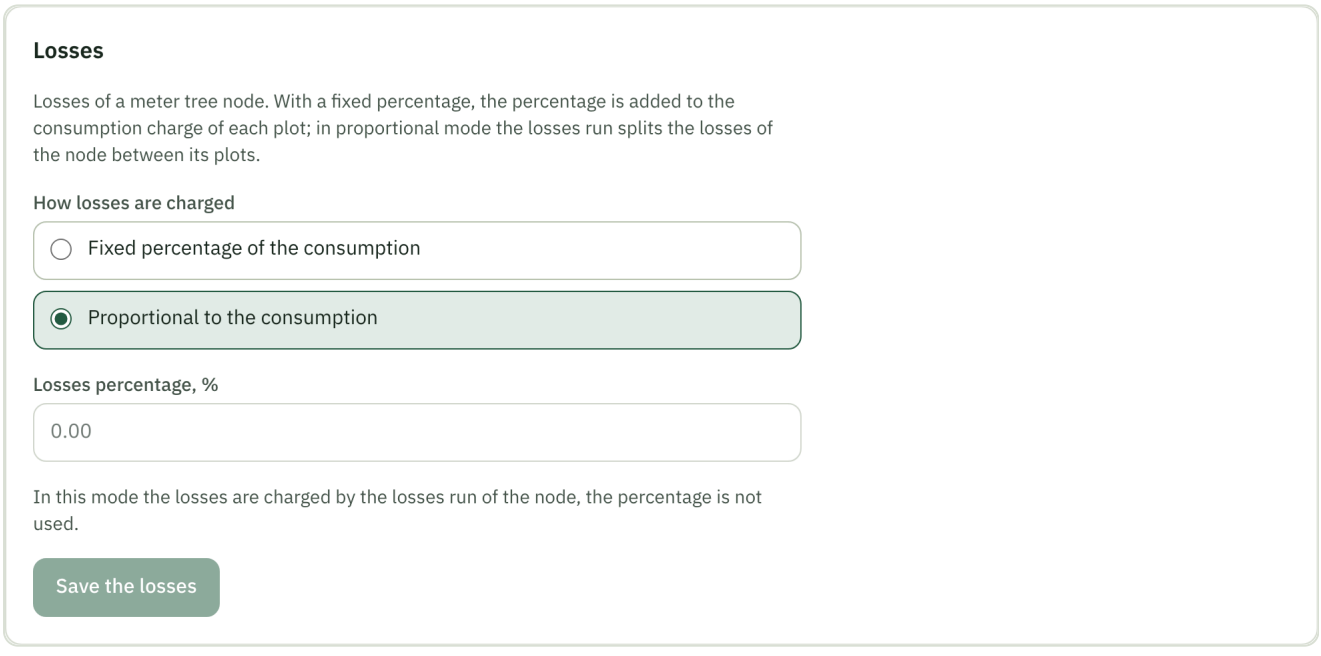


A losses preview: a node with an unknown value is skipped with a warning

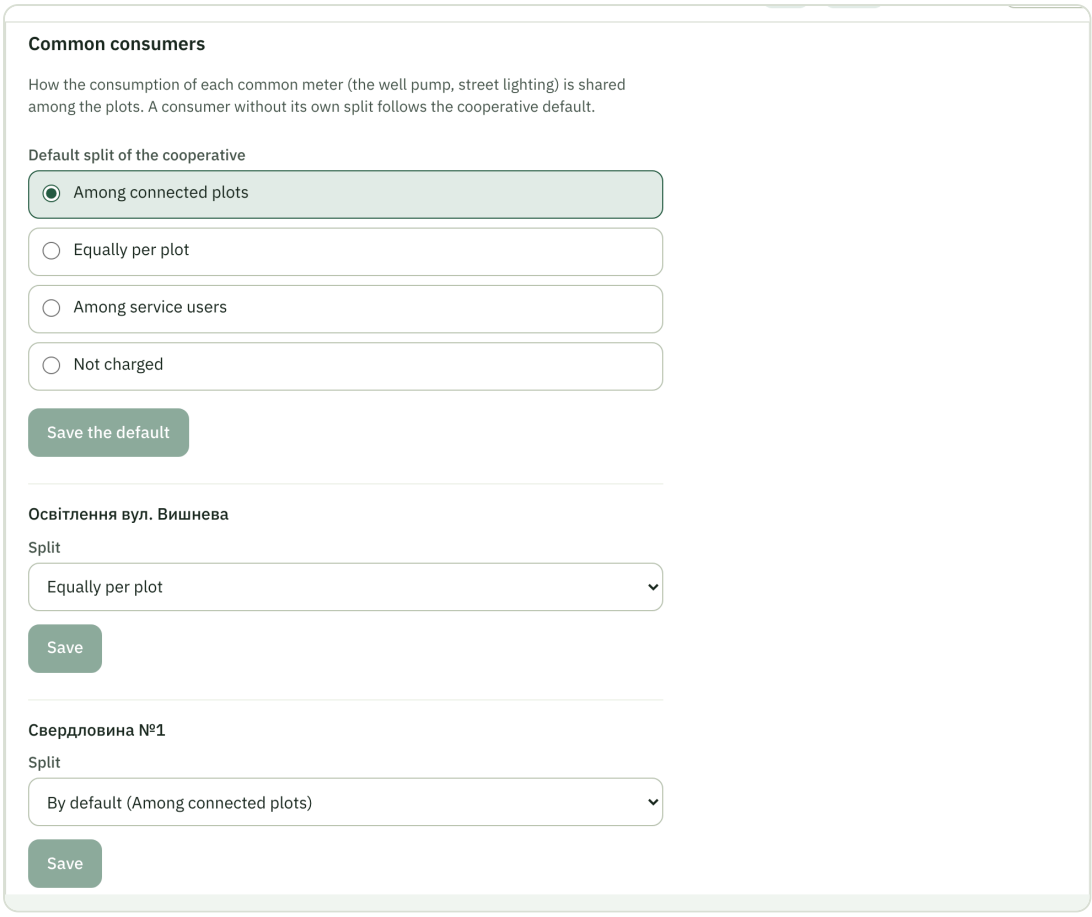
- The losses mode is chosen in the settings ("Losses"): with a fixed percentage there is nothing to run here and the page says so.
- If some nodes were already charged for these days the whole run cannot be booked; "Preview without the charged nodes" books the rest.
- A node with an unknown value goes to "Not charged"; when readings arrive, "Retry" charges it alone.
- Common consumers need a tariff of the "Common consumers" charge type; without it the preview says "No tariff" and nothing can be booked. The split of each consumer, its service users and the default are in the settings.



A common consumers preview



The losses mode in the settings



The split of each common consumer

19 Water and connections works

Water works like electricity: **a plot not connected to the well pays nothing for water**.

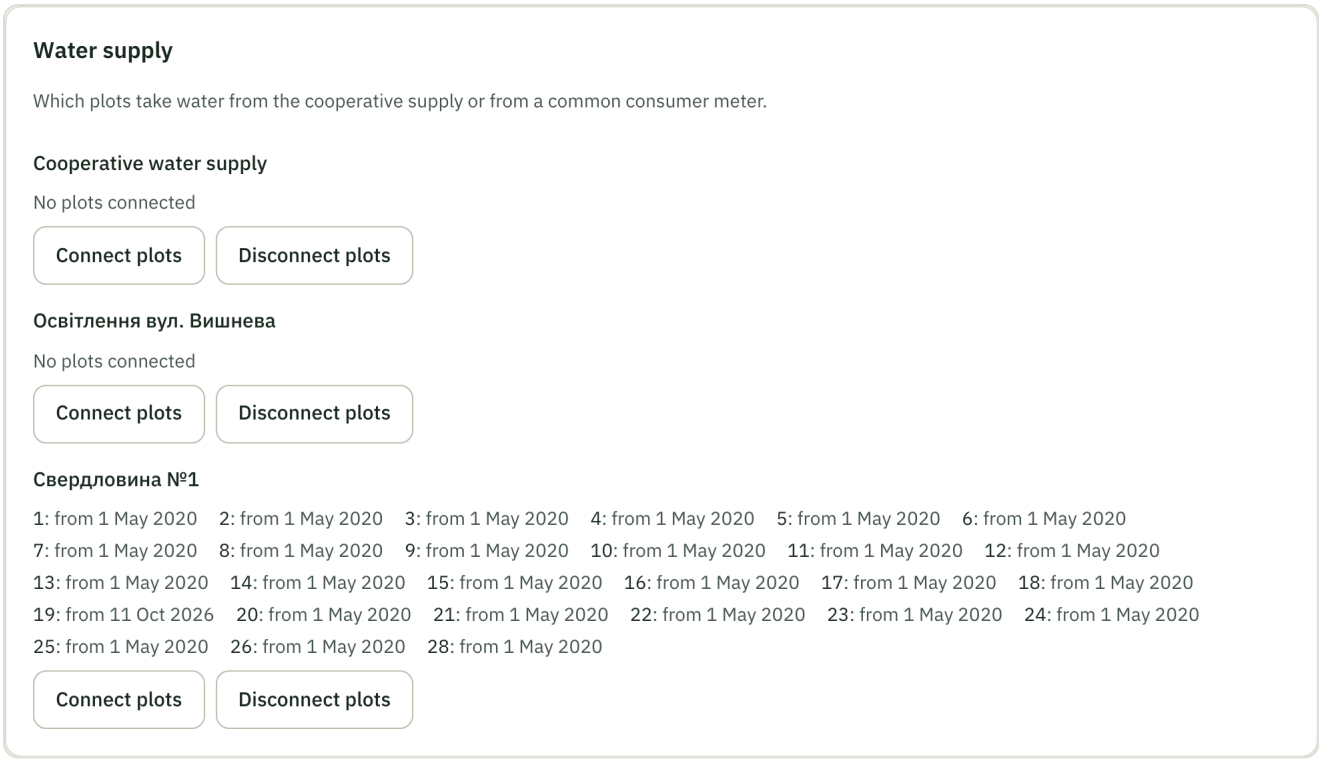
- There may be several wells: each plot is connected to a particular well, and each well's pump is shared only among its own plots.
- A plot has a history of water connections: the source (a well or "the cooperative's water supply"), from a date, until a date.
- Any water tariff — by water meter (m³) or fixed (per plot per season, say) — is charged only for the connected days. A per-m³ tariff also needs a water meter.
- A connection starting or ending mid-month splits the amount by days: 300 ₴ per year, connected from 16.10 (31 days) → 300 / 12 × 16 / 31 = **12.90 ₴**.
- The well pump's electricity is shared only among the plots connected to it, by plot-days: P1 and P2 the whole month, P3 from 16.10, pump 60 kWh → 30 + 30 + 15 = 75 plot-days → P1 24, P2 24, P3 12 kWh.
- Connections are entered on the plot card, in bulk on the well's page, or with the "Water" column of the import.

On the screen

A plot's connection is on its card, panel "Water": "Connect" (source and the first connected day) and "Disconnect" (the last connected day); the history is listed below. Many plots at once are connected or disconnected on the "Water supply" section of "Cooperative settings", per source. Without a connection a plot is never queued in "Not charged" for water and pays nothing for it.



The "Water" panel of a plot



The water sources and their plots in the settings

TWO THINGS TO KNOW

- If a well meter has been removed, its lasting connections are no longer offered in the settings: end them on each plot's card.
- Archiving a plot or making it common land does **not** end its lasting water connection. End the connection on the plot card first.

20 Corrections and recalculation

planned · 012

A back-dated tariff version, a correction of a used version or of a used reading — one mechanism:

- **Open month**: the old charge is reversed and a new one created.
- **Closed month**: nothing in it changes; the current open month gets an adjustment for the difference, referring to the original month.
- A reason is required. Before posting, a preview: how many plots and how much.
- If a debt goes down after it was already paid, the excess becomes a prepayment, never a "negative debt".

EXAMPLE

From 1 September 60 ₴/sotka/year was entered by mistake instead of 50; a 6-sotka plot. September is closed, October open.

	Was	After the correction
September (closed)	30.00	Unchanged; in October an adjustment −5.00 "for September"
October (open)	30.00	Reversal −30.00 and a new charge 25.00
Total	60.00	50.00; if 60.00 was paid, a prepayment of 10.00

21 Reports and templates planned · 013

- Built-in reports with filters and Excel download: charges for a period, plot balances, debtors, electricity by the tree (readings, cost per kWh, amount, losses in kWh and %), missing readings, land tax.
- **Templates**: the chairman uploads their own Excel file with placeholders such as `${cooperative.name}`, `${period.from}` and rows `${rows.plots.number}`; the system fills it in. The catalogue of all placeholders is on the templates page.
- **Tax report**: for now a placeholder "template not added yet"; the official form is added later as a template, without code changes.
- The cooperative's requisites join the settings: full name, ЄДРПОУ code, address, bank, IBAN.

22 What comes next planned

What	What it gives the board
Payments	Entering payments, a cash desk with receipts, bank statement import, allocation, prepayments.
Invoices and delivery	PDF invoices with the NBU QR code, no-debt certificates, delivery by email, Telegram, Viber.
Mobile app and map	An app for members and the electrician with photo readings; the cooperative map coloured by debt.

This guide is updated with every new capability of the system.